

AN EXPLORATION OF THE PARTIAL RESPECTS IN WHICH AN AXIOM SYSTEM RECOGNIZING SOLELY ADDITION AS A TOTAL FUNCTION CAN VERIFY ITS OWN CONSISTENCY

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Abstract. This article will study a class of deduction systems that allow for a limited use of the modus ponens method of deduction. We will show that it is possible to devise axiom systems α that can recognize their consistency under a deduction system D provided that: 1) α treats multiplication as a 3-way relation (rather than as a total function), and that 2) D does not allow for the use of a modus ponens methodology above essentially the levels of Π_1 and Σ_1 formulae.

Part of what will make this boundary-case exception to the Second Incompleteness Theorem interesting is that we will also characterize generalizations of the Second Incompleteness Theorem that take force when we only slightly weaken the assumptions of our boundary-case exceptions in any of several further directions.

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§1. Introduction. Let $A(x, y, z)$ denote a 3-way predicate relation indicating that $x + y = z$, and let $M(x, y, z)$ indicate that $x * y = z$. Let us say an axiom system α recognizes addition and multiplication as “**Total**” Functions iff it can prove:

$$(1) \quad \forall x \forall y \exists z A(x, y, z) \text{ AND } \forall x \forall y \exists z M(x, y, z)$$

Section 2 of this article will define a “U-Grounding Language” that formally recognizes addition as a total function, but treats multiplication as only a 3-way relation. Section 2 will use the terminology Σ_n^* and Π_n^* to describe the analogs of classical arithmetic’s Σ_n and Π_n sentences within this U-Grounding language notation. Thus, Section 2’s Σ_n^* and Π_n^* sentences will differ from arithmetic’s classic Σ_n and Π_n sentences by omitting a notational assumption, whose connotation would cause multiplication to be automatically treated as an implicitly defined total function.

The term **K–Deduction** will refer to an inference methodology, defined in Section 2, that contains a modus ponens rule for Σ_K^* and Π_K^* sentences (but not for higher sentences). The main result of this paper will be that it is feasible to devise a class of axiom systems for a U-Grounding language that recognize their own self-consistencies relative to 1–deduction.

Our result about 1-deduction will be near maximal because 1) an accompanying paper [70] has shown that a 2-deductive analog of our self-justifying formalism is provably infeasible, and because 2) Section 6 will introduce a new negativistic result that illustrates another direction in which our formalism cannot be further extended.

Moreover our new $IS_D(A)$ formalism is stronger than its predecessor, called $IS(A)$ in [67], and $IS_D(A)$'s associated proof is also significantly simpler.

Underlying Formalism: Some notation is needed to start our discussion. Let α denote an axiom system and D denote a deduction method. The pair (α, D) will be called an **Introspectively Unified Logic** iff

- i: one of α 's formal theorems will state that the deduction method D , applied to the axiom system α , will produce a consistent set of theorems,
- ii: and the axiom system α is in fact consistent.

Also, an axiom system α will be called **Self-Justifying** iff there exists some frequently employed deduction method D , such as perhaps the Semantic Tableaux, Hilbert, Herbrand or Sequent-Calculus formalisms, where (α, D) is an Introspectively Unified Logic.

Kleene's Fixed Point Theorem implies that every r.e. axiom system α can be easily extended into a broader system α^* which satisfies condition (i). For a fixed deduction method D , Kleene proposed [24] to let α^* contain all α 's axioms plus one added axiom sentence stating :

+ There is no proof (using deduction method D) of a prototypical absurdity-sentence (such as for example " $0=1$ ") from the union of the axiom system α with *this* sentence "+" (looking at itself).

Kleene explained how to apply the Fixed Point Theorem to encode a self-referencing statement, similar to the axiom (above). However, he pointed out that the catch is that α^* may be inconsistent even while its added axiom formally asserts α^* 's consistency. For this reason, Kleene, Rogers and Jeroslow [22, 24, 46] each warned their readers that most axiom systems, similar to α^* , were useless on account of their inconsistency, *although they were technically well-defined*. In the nomenclature of the previous paragraph, the difficulty is that (α^*, D) may satisfy Part-i but not the equally important Part-ii of the definition of introspective unification.

This problem arises in settings considerably more general than Gödel's original paradigm, where α was an extension of Peano Arithmetic. For instance, Section 4's literature survey will discuss a large class of more general settings [1, 2, 3, 4, 7, 9, 14, 20, 23, 37, 38, 40, 41, 42, 43, 49, 52, 53, 57, 58, 64, 68, 70], where sundry variations of Gödel's Second Incompleteness Theorem apply to weak axiom systems. Each of these generalizations of the Second Incompleteness Theorem can be mapped onto a paradigm where it is impossible to construct a self-justifying formalism (α, D) .

Our interest in this topic began with the observation that there are certain settings where it is actually feasible to construct introspectively unified logics (α, D) . The two main examples discussed in our previous work were the particular cases where:

- A:** The axiom system α recognized addition *but not* multiplication as a total function, and D was some cut-free deductive method, such as semantic tableaux or Herbrand deduction. [67].
- B:** The system α did not assume that either addition or multiplication was total, but now D represented Hilbert deduction. (The strongest version of this paradigm appeared in [72], where an “additive naming convention” had replaced the conventional axiom declaring addition is total.)

Items (A) and (B) naturally raise the question whether it is feasible to hybridize these effects. One of our results will show that some types of hybridizations are possible, under a formalism called $IS_D(A)$, which recognizes addition as a total function and uses a deductive method that allows for the application of a modus ponens principle for Π_1^* and Σ_1^* sentences.

Part of what will make $IS_D(A)$ interesting is that there are several generalizations of Gödel’s Second Incompleteness Theorem, that indicate $IS_D(A)$ is difficult to further improve. Among these generalizations of the Second Incompleteness are the following phenomena:

1. Pudlák [41] has demonstrated that no extension of axiom system Q [7, 58] can verify its own Hilbert consistency localized on a definable cut. Solovay [52] has revised Pudlák’s formalism with some additional techniques of Nelson and Wilkie-Paris [34, 64] to establish that essentially all axiom systems recognizing Successor as a total function are unable to verify their own Hilbert consistency. (Section 4 will review this prior literature in considerable detail.)
2. We developed in [70] a generalization of Item 1’s result which showed that all axiom systems recognizing addition as a total function are unable to verify their 2-deductive consistency.
3. Expanding upon some techniques pioneered by Adamowicz, Paris, Pudlák, Solovay, Takeuti, Wilkie and Zbierski in [1, 4, 38, 41, 52, 57, 64], we developed in [68] a further new result which showed that no natural axiom system can recognize multiplication as total and also verify its cut-free consistency (or “semantic tableaux” [66] consistency).
4. Section 6 of our current article will describe a close cousin of $IS_D(A)$, called $NS_D^{k,m}(A)$, that is essentially always inconsistent. This fact will be interesting because the only substantial difference between $IS_D(A)$ and $NS_D^{k,m}(A)$ will be the respective manners in which they define the word “this” in the Statement $+$ ’s Kleene-like axiom.

A detailed review of the prior literature will appear in Section 4. It will explain more fully exactly how our work is related to the prior literature. Our two new results about

$IS_D(A)$ and $NS_D^{k,m}(A)$ shall represent a pair of closely matching positive and negative results. Also, $IS_D(A)$ will be significant because it will fill most of the gap that was left unexplored by results (1)–(4).

In essence, the close match between these positive and negative results will be mathematically significant, regardless of whether or not it is found later to have added significance for epistemology. From a mathematical standpoint, these theorems are useful because the mixture of their opposing positive and negative results will help characterize the exact threshold-level where the Second Incompleteness Theorem takes force.

This paper will be divided into three parts. Sections 2 and 3 will summarize our new results. Section 4 will be a literature survey. Sections 5 and 6 will prove our main new theorems.

This article is entirely self-contained. It should be comprehensible to most logicians. A researcher who has taken an *intensive* 2-semester graduate-level course on logic should probably be able to follow most of the gist of this paper.

§2. Description of Main Results. Let us define a mapping $F(a_1, a_2, \dots, a_j)$ to be a **Non-Growth** function iff $F(a_1, a_2, \dots, a_j) \leq \text{Maximum}(a_1, a_2, \dots, a_j)$ for all values of $a_1, a_2, \dots, a_j$. Six examples of non-growth functions are:

1. *Integer Subtraction* where $x - y$ is defined to equal zero when $x \leq y$,
2. *Integer Division* where $x \div y$ equals x when $y = 0$, and it equals $\lfloor x/y \rfloor$ otherwise,
3. *Root*(x, y) which equals $\lceil x^{1/y} \rceil$ when $y \geq 1$, and it equals x when $y = 0$.
4. *Maximum*(x, y),
5. *Logarithm*(x) = $\lfloor \text{Log}_2(x) \rfloor$ when $x \geq 2$, and it equals zero otherwise.
6. *Count*(x, j) designating the number of “1” bits among x ’s rightmost j bits.

These six function are called the **Grounding Functions**. Also, the term **U-Grounding Function** will henceforth refer to a set of functions that includes these six operations plus the growth functions of addition and $\text{Double}(x) = x + x$.

All our results will use a U-Grounding language. This notation is technically unnecessary — because a system that uses the combination of Equation (2) (which implies addition and doubling are total functions) along with only the first six non-growth Grounding operations would have properties similar to the U-Grounding language.

$$(2) \quad \forall x \forall y \exists z \ x = z - y$$

However, our notation is much simplified under a language employing function symbols also for addition and $\text{Double}(x) = x + x$. The virtue of this notation is that it uses no more than $2 \text{Log}_2 N$ appearances of the addition and doubling function symbols (applied to the input of “1”) to encode every integer $N \geq 2$. Such a term will be called a **U-Grounded Binary Representation of N**, and it will be denoted as $\overline{N}$. For

instance, $\overline{25}$ can be encoded as:

$$(3) \quad 1 + \text{Double}(\text{Double}(\text{Double}(1 + \text{Double}(1)))).$$

The use of logic's conventional notation about Π_n and Σ_m sentences is technically inappropriate in this paper because the latter notation (with its multiplication function symbol) is suitable only for axiom systems which recognize multiplication as a total function. Instead, our analogs for Π_n and Σ_m in the U-Grounding Function language are called Π_n^* and Σ_m^* . Here, a *term* t is defined to be a constant, variable or a U-Grounding function symbol (whose input arguments are recursively defined terms). Also, the quantifiers in the wffs $\forall v \leq t \Psi(v)$ and $\exists v \leq t \Psi(v)$ are called *bounded quantifiers*. We will say a formula Φ belongs to the Δ_0^* class if it uses the U-Grounding primitives as its function symbols, the two relation symbols of “=” and “<”, and all its quantifiers are bounded. (Following conventional logic notation, we will sometimes also use the terms “ Π_0^* ” and “ Σ_0^* ” to designate the Δ_0^* class.) For $n \geq 1$, a formula Υ shall be called Π_n^* iff it is written in the form $\forall v_1 \forall v_2 \dots \forall v_k \Phi$, where Φ is Σ_{n-1}^* . Likewise, Υ is called Σ_n^* iff it is written in the form $\exists v_1 \exists v_2 \dots \exists v_k \Phi$, where Φ is Π_{n-1}^* . Also, Υ is called Q_n^* iff it is a Boolean combination of several Σ_n^* and Π_n^* formulae (using the standard connective symbols of $\wedge$, $\vee$, $\neg$ and $\rightarrow$).

The preceding notation is the same as [20]'s quite conventional definitions for Σ_n , Π_n and Q_n except that the definitions of Σ_n^* , Π_n^* and Q_n^* employ a U-Grounding language instead of a language using the primitives of arithmetic. In particular since our U-Grounding language has no function symbol for multiplication, our U-Grounding based axiom systems α do not implicitly assume that multiplication is a total function. However, they do certainly retain an ability to encode the graph $M(x, y, z)$ of multiplication as a Δ_0^* formula. For instance, one possible Δ_0^* encoding for $M(x, y, z)$ is illustrated below:

$$(4) \quad [(x = 0 \vee y = 0) \Rightarrow z = 0] \wedge [(x \neq 0 \wedge y \neq 0) \Rightarrow (\frac{z}{x} = y \wedge \frac{z-1}{x} < y)]$$

Equation (4)'s formal encoding of $M(x, y, z)$'s graph as a Δ_0^* formula will add a substantial level of semantic depth to our U-Grounding axiom systems α because it will imply that these logics will have an understanding of multiplication's main generic properties (except of course for the *deliberately omitted* totality condition of “ $\forall x \forall y \exists z M(x, y, z)$ ”.)

In a context where (α, D) is an ordered pair with α representing an axiom system and with D designating a deduction method, we will also use the following notation:

1. A **Level(n) Definition** of (α, D) 's consistency is the declaration that there exists no Π_n^* sentence Υ such that (α, D) supports simultaneous proofs of both Υ and $\neg \Upsilon$.
2. A **Level(0-) Definition** of (α, D) 's consistency is the statement that there exists no proof of $0=1$ from the (α, D) configuration.

All the definitions of consistency, from Level(0-) up to Level(n) for any n , are equivalent to each other under strong enough models of arithmetic. However, many weak axiom systems do not have a mathematical strength to recognize this equivalence.

Our basic goal will be to determine for what configurations (α, D) is it possible to construct introspectively unified logics that have a Level(1) understanding of their own self-consistency. As we shall explain, it will turn out that it is impossible to construct self-justifying formalisms much above the Level(1). Indeed, there will be some respects in which a formalism is unable to have even a Level(0-) appreciation of its own self-consistency.

It is helpful to review the definition of a semantic tableaux proof from Fitting's or Smullyan's textbooks [13, 51] before summarizing our new results. Let us call a sentence **Prenex* Normalized** iff it is either Π_i^* or Σ_i^* sentence, for some $i \geq 0$. Define a **Φ -Focused Candidate Tree** for the axiom system α to be a tree structure whose root corresponds to the sentence $\neg \Phi$ rewritten in Prenex* form and whose all other nodes are either axioms of α or deductions from higher nodes of the tree. Let the notation " $\mathbf{A} \Rightarrow \mathbf{B}$ " indicate that $\mathbf{B}$ is a valid deduction when $\mathbf{A}$ is an ancestor of $\mathbf{B}$ in a specified proof tree. In this notation, the deduction rules allowed in a candidate tree are:

1. $\Upsilon \wedge \Gamma \Rightarrow \Upsilon$ and $\Upsilon \wedge \Gamma \Rightarrow \Gamma$.
2. $\neg \neg \Upsilon \Rightarrow \Upsilon$. Other rules for the " $\neg$ " symbol include: $\neg(\Upsilon \vee \Gamma) \Rightarrow \neg \Upsilon \wedge \neg \Gamma$, $\neg(\Upsilon \rightarrow \Gamma) \Rightarrow \Upsilon \wedge \neg \Gamma$, $\neg(\Upsilon \wedge \Gamma) \Rightarrow \neg \Upsilon \vee \neg \Gamma$, $\neg \exists v \Upsilon(v) \Rightarrow \forall v \neg \Upsilon(v)$ and $\neg \forall v \Upsilon(v) \Rightarrow \exists v \neg \Upsilon(v)$.
3. A pair of sibling nodes Υ and Γ is allowed when their ancestor is $\Upsilon \vee \Gamma$.
4. A pair of sibling nodes $\neg \Upsilon$ and Γ is allowed when their ancestor is $\Upsilon \rightarrow \Gamma$.
5. $\exists v \Upsilon(v) \Rightarrow \Upsilon(u)$ where u is a newly introduced "Parameter Symbol".
6. Our variation of Rule 5 for *bounded existential quantifiers* of the form " $\exists v \leq s$ " is the identity: $\exists v \leq s \Upsilon(v) \Rightarrow u \leq s \wedge \Upsilon(u)$.
7. $\forall v \Upsilon(v) \Rightarrow \Upsilon(t)$ where t denotes a U-Grounded term. These terms are defined to be parameter symbols, constant symbols, or U-Grounding functions with recursively defined inputs. (Without loss of generality, we may assume that each "parameter symbol" u , appearing in such a term t , is required to be introduced by some ancestor of the current node, which has used one of the Rules 5 or 6.)
8. Our variation of Rule 7 for *bounded universal quantifiers* of the form " $\forall v \leq s$ " is the identity: $\forall v \leq s \Upsilon(v) \Rightarrow t \leq s \rightarrow \Upsilon(t)$.

Define a particular leaf-to-root branch in a candidate tree T to be **Closed** iff it contains both some sentence Υ and its negation $\neg \Upsilon$. A **Semantic Tableaux** proof of Φ is then defined to be a candidate tree, *all of whose root-to-leaf branches are closed*, such that the tree's root stores the sentence $\neg \Phi$ (rewritten in Prenex* normal form) and all its other nodes are either axioms of α or deductions from higher nodes.

The only distinction between our definition of a semantic tableaux proof and some other conventional definitions in [13, 51, 64] is that we require Φ 's proof tree to have

its root store $\neg \Phi$ written in *Prenex* normal form*, whereas some other conventional definitions do not have the Prenex* requirement. All our results will also hold if we drop the Prenex* requirement. However, the notation will be greatly simplified if we begin with the Prenex* assumption.

We are now ready to define the main deductive method studied in this paper. Let H denote a sequence of ordered pairs $(t_1, p_1), (t_2, p_2), \dots (t_n, p_n)$, where p_i is a Semantic Tableaux proof of the theorem t_i , and where $\mathfrak{R}$ is an arbitrary class of sentences. Define H to be a Tab- $\mathfrak{R}$ -List proof of a theorem T from the axiom system α iff $T = t_n$ and also:

1. Each axiom in p_i 's proof is either one of $t_1, t_2, \dots, t_{i-1}$ or comes from α .
2. Each of the “intermediately derived theorems” $t_1, t_2, \dots, t_{n-1}$ must lie within the “pre-specified class” $\mathfrak{R}$ of sentences.

Tab- $\mathfrak{R}$ -List proofs are thus a stronger deductive method than the semantic tableaux by allowing for a limited type of modus ponens rule for sentences that belong to the intermediate class, that is formalized by $\mathfrak{R}$. For instance, if $\mathfrak{R}$ represents the set of Q_K^* sentences, then a Tab- $\mathfrak{R}$ -List proof would represent a modified form of semantic tableaux that allows for modus ponens for the Q_K^* class of sentences. Henceforth, this style of proof will be called a “Tab- Q_K^* -List” proof. (If $\mathfrak{R}$ represents the universal set of all sentences, then Tab- $\mathfrak{R}$ -List deduction will have a deductive-efficiency essentially equivalent to that of a Hilbert-style proof.)

There are several forms of tab-list deduction that lie between Tab- Q_{k-1}^* and Tab- Q_k^* List deduction. For instance, let U_k^* denote the union of the Σ_k^* and Π_k^* classes. Then a Tab- U_k^* -List proof will be defined as a variant of tab-list deduction where $\mathfrak{R}$ represents the set of U_k^* sentences. Similarly, the Tab- Σ_k^* and Tab- Π_k^* -List deductive methods will be variants of the tab-list formalism where $\mathfrak{R}$ represents the class of Σ_k^* and Π_k^* sentences (respectively).

This notation is useful for describing those introspectively unified logics that are feasible to construct, as well as those which are infeasible. Let A denote an arbitrary consistent axiom system that employs the U-Grounding language. Then our main positive result will be that if D represents either the Tab- U_1^* -List (or yet the stronger Tab- Q_1^* deductive method), then it is possible to construct an introspectively unified logic (α, D) where α is an axiom system capable of recognizing addition as a total function and also of verifying all A 's Π_1^* theorems. (Both such (α, D) will recognize their self-consistency under a Level(1) definition for (α, D) -styled consistency.)

On the other hand, a complementary negative result from [70] shows that no analog of this effect is available for the Tab- Σ_2^* and Tab- Π_2^* -List deductive methods — *even* when one uses the *fully weak Level(0-)* definition of self-consistency. Moreover, there are a large number of other types of generalizations of the Second Incompleteness Theorem, delineating other respects in which it is impossible to further expand or improve upon Section 3's proposed $IS_D(A)$ formalism.

One of these negativistic results will appear in Section 6 of the current paper: It will demonstrate some inherent limitations associated with Kleene-like “I am consistent” axioms.

Other directions in which it is impossible to further improve upon $IS_D(A)$ will be discussed during Section 4’s literature survey.

§3. Formal Description of the $IS_D(A)$ and $NS_D^{k,m}(A)$ Axiom Systems. Let A denote any axiom system which employs the U-Grounding language, and D denote a deduction method. The symbol $IS_D(A)$ will denote an axiom system, associated with the ordered pair (A, D) , that has a simultaneous capacity to:

1. prove all A ’s Π_1^* theorems,
2. recognize addition as a total function, and
3. contain a Kleene-like self-referencing axiom asserting its own Level(1) consistency under the deduction method D .

This section will define the $IS_D(A)$ axiom system and its similar-looking cousin, called $NS_D^{k,m}(A)$. Despite their superficial resemblances, these two axiom system will actually have nearly the opposite properties. Our main goal will be to determine: *Which deduction methods D shall make the $IS_D(A)$ formalism consistent?* It will be a consequence of the joint research of largely Pudlák and Solovay (reviewed during Section 4’s literature survey) that if D denotes Hilbert deduction then our $IS_D(A)$ system will be automatically inconsistent.

Formal Definition of $IS_D(A)$: In a context where A is any axiom system employing the U-Grounding language and D denotes a deduction method, $IS_D(A)$ is defined to be an axiom system with the following four sub-groups of axioms:

Group-Zero: Two of the Group-zero axioms will define the two initial named constant symbols, $\bar{c}_0$ and $\bar{c}_1$, that designate the integers of 0 and 1. The third and fourth Group-zero axioms will define the growth functions of addition and $Double(x) = x + x$. In conjunction, the net effect of these four Group-zero axioms will be to set up the machinery to define unambiguously any natural number $n \geq 2$ as a term using only the constant symbol $\bar{c}_1$ and employing $O(\log n)$ appearances of the function symbols for the addition and doubling operations. Section 2 had indicated that such a term, analogous to Equation (3), which encodes n in a binary-like format, will be denoted as $\bar{n}$.

Group-1: This axiom group will consist of a finite set of Π_1^* sentences, called F , such that the union of F with the Group-zero axioms is sufficient to prove every Δ_0^* sentence that is valid under the standard model of the natural numbers. (Any finite set of Π_1^* sentences F with this property may be used to define Group-1. One suitable finite set of axioms was defined by Table I of our article [67]. However, any other alternate finite set of Π_1^* axioms is equally suitable.)

Group-2: Throughout this paper, $\lceil \Phi \rceil$ will denote the Gödel number of Φ . Also $\text{Prf}_A(t, p)$ will denote a Δ_0^* formula indicating that p is a proof of the theorem t from the axiom system A . For each Π_1^* sentence Φ , the Group-2 schema will contain an axiom of the form (5), written in Π_1^* normalized form. Thus, the Group-2 scheme shall trivially endow $\text{IS}_D(A)$ with a capacity to verify all A 's Π_1^* theorems.

$$(5) \quad \forall p \{ \text{Prf}_A(\lceil \Phi \rceil, p) \rightarrow \Phi \}$$

Group-3: This axiom group will consist of one single Π_1^* sentence stating essentially that:

- * In the context of the deduction methodology D , there exists no two proofs of both some Π_1^* sentence and its formal negation from the union of the Group-zero, 1 and 2 axioms with **this sentence (itself)**.

In order to formally encode the self-referential statement $*$ as a Π_1^* sentence, let $\text{Pair}(x, y)$ denote a Δ_0^* formula indicating x is the Gödel number of a Π_1^* sentence and y is x 's negation. Also, let $\text{Prf}_{\text{IS}_D(A)}(t, b)$ denote a Δ_0^* formula that indicates that b is a proof (using the deduction method D) of the theorem t from the axiom system $\text{IS}_D(A)$. In this context, the formal encoding of $*$ can be approximated as:

$$(6) \quad \forall x \forall y \forall p \forall q \neg [\text{Pair}(x, y) \wedge \text{Prf}_{\text{IS}_D(A)}(x, p) \wedge \text{Prf}_{\text{IS}_D(A)}(y, q)]$$

More Detailed Description of the Group-3 Axiom: A formal description of $\text{IS}_D(A)$'s Group-3 axiom is more complicated than the abbreviated descriptions given either by Sentence $*$ or by Equation (6)'s analog. The main added complication is because the Group-3 axiom declares the consistency of a formal set of axioms that includes “itself” (in the words of Sentence $*$). As was noted in Section 1, the notion of an axiom including “itself” when it refers to the consistency of an axiom schema dates back to Kleene's 1938 paper [24]. However, Kleene's abbreviated description is insufficient to establish that Equation (6) can be encoded precisely as a Π_1^* sentence. The next two paragraphs will explain how this can be done.

Let $\text{UNION}(A)$ denote the union of $\text{IS}_D(A)$'s Group-Zero, Group-1 and Group-2 axioms. We will also use the following notation:

- i: $\text{Prf}_{\text{UNION}(A)}^D(t, p)$ will denote a formula designating that p is a proof of the theorem t from the axiom system $\text{UNION}(A)$ using the deduction method D .
- ii: $\text{ExPrf}_{\text{UNION}(A)}^D(h, t, p)$ will be a formula stating that p is a proof (using the deduction method D) of a theorem t from the union of the axiom system $\text{UNION}(A)$ with the added axiom sentence whose Gödel number equals h .
- iii: $\text{Subst}(g, h)$ will denote Gödel's classic substitution formula — which yields TRUE when g is an encoding of a formula and h is an encoding of a sentence that replaces all occurrence of free variables in g with the formally Gödel-encoded term of $\overline{g}$.

- iv: $\text{SubstPrf}_{UNION(A)}^D(g, t, p)$ will denote the natural hybridizations of the constructs from Items (ii) and (iii) which yields a Boolean value of TRUE exactly when there exists some integer h simultaneously satisfying *both* the conditions $\text{Subst}(g, h)$ and $\text{ExPrf}_{UNION(A)}^D(h, t, p)$.

Each of (i)–(iv) can be encoded as Δ_0^* formulae. Thus, Appendices C and D of [67] explained how the first three of these predicates can receive Δ_0^* encodings when one applies the theory of LinH functions [20, 25, 73]. In such a context, Equation (7) illustrates one possible Δ_0^* encoding for $\text{SubstPrf}_{UNION(A)}^D(g, t, p)$ ’s graph. (It is equivalent to “ $\exists h [\text{Subst}(g, h) \wedge \text{ExPrf}_{UNION(A)}^D(h, t, p)]$ ”, but Equation (7) is a Δ_0^* formula — *unlike* the quoted expression.)

(7)

$$\text{Prf}_{UNION(A)}^D(t, p) \vee \exists h \leq p [\text{Subst}(g, h) \wedge \text{ExPrf}_{UNION(A)}^D(h, t, p)]$$

Utilizing (7)’s Δ_0^* encoding for $\text{SubstPrf}_{UNION(A)}^D(g, t, p)$, it is easy to encode the Group-3 axiom of $\text{IS}_D(A)$ as a Π_1^* sentence. Thus, let $\Gamma(g)$ denote Equation (8)’s formula, and let n denote $\Gamma(g)$ ’s Gödel number. Then “ $\Gamma(\overline{n})$ ” is a Π_1^* sentence encoding this Group-3 axiom.

(8)

$$\forall x \forall y \forall p \forall q \neg [\text{Pair}(x, y) \wedge \text{SubstPrf}_{UNION(A)}^D(g, x, p) \wedge \text{SubstPrf}_{UNION(A)}^D(g, y, q)]$$

Thus in a context where n denotes (8)’s Gödel number, the construct $\text{Prf}_{\text{IS}_D(A)}(t, p)$ can receive a formal Δ_0^* encoding as : “ $\text{SubstPrf}_{UNION(A)}^D(\overline{n}, t, p)$ ”.

Section 5 will use the preceding paragraph’s Δ_0^* encoding for “ $\text{Prf}_{\text{IS}_D(A)}(t, p)$ ” to interpret Equation (6)’s Group-3 axiom as a Π_1^* sentence. (This Π_1^* encoding will be important in Section 5. Our results would be substantially less significant if Equation (6) had a weaker Π_2^* encoding.)

The Broader Perspective of this Article: The potential difficulty pertaining to $\text{IS}_D(A)$ is that there is no guarantee that it is consistent. Thus using Section 1’s nomenclature, $\text{IS}_D(A)$ can potentially violate “Part-ii” of the definition for “introspective unification”

For instance, Section 4’s literature survey will explain how the combined work of Nelson, Pudlák, Solovay and Wilkie-Paris [34, 41, 52, 64] implies no consistent axiom system can recognize its own Hilbert-consistency when it recognizes Successor as total. Thus, $\text{IS}_D(A)$ can not satisfy Part-ii of the definition of introspective unification when D represents Hilbert deduction.

What types of deduction methods D do have the property that $\text{IS}_D(A)$ can satisfy Part-ii of the definition of introspective unification ? Three definitions will help answer this question.

DEFINITION 1. Consider a function $\mathfrak{S}$ which maps an initial axiom system A (employing the language of the U-Grounding functions) onto a second axiom system,

denoted as $\mathfrak{S}(A)$. Let us call this function $\mathfrak{S}$ **Consistency Preserving** iff $\mathfrak{S}(A)$ is guaranteed to be consistent whenever A 's Π_1^* theorems are valid in the standard model. (In this notation, the main problem considered in this article is to identify the axiom mapping functions $\mathfrak{S}(\bullet)$ that are consistency-preserving.)

DEFINITION 2. Throughout this paper, $\text{Log}(z)$ will denote Base-2 Logarithm, with *downwards rounding* to the lowest integer, and $\text{Log}^k(z)$ will denote $\text{Log}(\text{Log} \dots (\text{Log}(z)))$ — where k is an integer indicating the number of iterations of Log here. The symbol 2_k^m will denote the least integer z such that $\text{Log}^k(z) \geq m$. Also, $\Upsilon(k, m)$ will denote the Σ_1^* sentence defined below:

$$(9) \quad \exists z \text{Log}^k(z) \geq \overline{m}$$

DEFINITION 3. The axiom system $\text{NS}_D^{k,m}(A)$ shall have an identical definition as $\text{IS}_D(A)$, except for the following two changes:

- a: The Group-zero schema of $\text{NS}_D^{k,m}(A)$ will include $\Upsilon(k, m)$ as an added axiom.
- b: $\text{NS}_D^{k,m}(A)$'s Group-3 axiom will have an identical structure as $\text{IS}_D(A)$'s Group-3 axiom — except that $\text{NS}_D^{k,m}(A)$'s Group-3 axiom will be sensitive to the change that was made by Item (a) above. Thus, the meaning of the word “this” in the Group-3 axiom that declares “*This system is consistent*” will be sensitive to the fact that the construct “this” now refers to a system *that includes* $\Upsilon(k, m)$ *as an axiom under its scope*.

The two main theorems in this paper are Theorems 5 and 6. The former result will establish that $\text{IS}_D(\bullet)$ is a consistency-preserving transformation for the case where D denotes $\text{Tab}-U_1^*$ -List deduction. On the other hand, Theorem 6 will show that this consistency preservation property does not extend to $\text{NS}_D^{k,m}(\bullet)$ for most values of k and m — *even in the extremal case* where D represents a cut-free deductive method, such as semantic tableaux.

The contrast between Theorems 5 and 6 is clearly counter-intuitive. After all, the version of $\text{IS}_D(A)$ — where D represents $\text{Tab}-U_1^*$ -List deduction — has an ability to prove the validity of Equation (9)'s sentence $\Upsilon(k, m)$ and then to use this sentence as an intermediate step during the proofs of subsequent theorems.

However, such proofs would be sharply shorter if $\text{IS}_D(A)$ could treat $\Upsilon(k, m)$ as a starting axiom, similar to the $\text{NS}_D^{k,m}(A)$ formalism. In essence, this fact combined with Part-b of the definition of $\text{NS}_D^{k,m}(A)$ will alter sufficiently the meaning of the word “this” in Definition 3's formalization of $\text{NS}_D^{k,m}(A)$ for Theorems 5 and 6 to provide contrasting results.

One reason this topic is interesting is because it clarifies the nature of the *threshold* where the Second Incompleteness Theorem takes force. A second interesting aspect of

Theorem 5 is that it is stronger than [67]’s prior results because 1) it uses a Level(1) (rather than Level(0-)) definition of consistency and 2) its modus ponens rule extends to all Σ_1^* and Π_1^* sentences. Moreover, Section 4’s literature survey will explain that $IS_D(A)$ is near-maximal because a variety of generalizations of the Second Incompleteness Theorem show that it is difficult to further improve in most directions that one would want to pursue.

§4. Literature Survey. Research into the incompleteness phenomena began with Gödel’s 1931 paper [16] and with Rosser’s removal [47] of Gödel’s ω -consistency assumption. Gödel’s Second Incompleteness Theorem indicated that no extension of Peano Arithmetic can verify its own consistency. The literature has subsequently used the symbol Q to denote an axiom system weaker than Peano Arithmetic, which contains no induction schema and which essentially only recognizes addition and multiplication as total functions. Tarski-Mostowski-Robinson [58], showed a construct called “Essential Undecidability”, was valid for Q , all its extensions, but none of its subsets.

Several articles have subsequently developed formal generalizations of the Second Incompleteness Theorem for Q . The first of these, by Bezboruah-Shepherdson [7], explored a paradigm that was dependent on the exact encoding of the provability predicate. Stronger results were subsequently derived by other authors. Some further notation is needed to summarize these results.

Let $S(x)$ denote the “successor” operation that maps the integer x onto $x + 1$. A formula $\varphi(x)$ is called [20] a **Definable Cut** for an axiom system α iff α can prove:

$$(10) \quad \varphi(0) \text{ AND } \forall x \{ \varphi(x) \rightarrow \varphi[S(x)] \} \text{ AND } \forall x \forall y < x \{ \varphi(x) \rightarrow \varphi(y) \}$$

Definable cuts have been studied by a very extensive literature [1, 4, 9, 10, 17, 19, 20, 23, 26, 25, 34, 37, 38, 40, 41, 43, 55, 60, 61, 63, 64]. All axioms systems, strictly weaker than Peano Arithmetic, contain some definable cut that is not provably equivalent to the full set of integers. (In the proof-theory literature, the definition of “Definable cuts” is conceptually unrelated to Gentzen’s notion of a sequent calculus deductive “cut rule”, despite their similar sounding names.)

In a context where $\Upsilon(x)$ and $\varphi(x)$ denote two definable cuts, the formula $\varphi(x)$ has been called a **Thinning** of $\Upsilon(x)$ (relative to α) iff α can prove:

$$(11) \quad \forall x [\varphi(x) \rightarrow \Upsilon(x)]$$

Several articles [9, 20, 23, 25, 34, 37, 41, 43, 44, 61, 64, 67] have credited some unpublished comments of Robert Solovay for observing (in Definition 2’s notation) that for each integer K and definable cut $\Upsilon(x)$, there exists a thinning $\varphi(x)$ of $\Upsilon(x)$ where *most naturally-formed* arithmetic axiom system α can prove:

$$(12) \quad \forall p [\varphi(p) \rightarrow \exists q \Upsilon(q) \wedge q = 2_K^p] \text{ AND } \forall p [\varphi(p) \rightarrow \exists r \varphi(r) \wedge r = 2_K^{2 \cdot \text{Log}^K p}]$$

It is somewhat awkward to provide an exact summary of Solovay's theorem because Solovay never published it, and several authors [9, 20, 23, 25, 34, 37, 41, 43, 44, 61, 64, 67] have provided several slightly different interpretations of it. For instance, we were deliberately vague in the second sentence of this paragraph when stating Solovay's theorem applied to "*most naturally-formed*" arithmetic systems, since the latter italicized phrase can have several different interpretations. One possible resolution for this ambiguity is to introduce the notion of a **Successor-based Arithmetic** axiom system, as formally defined in the Footnote ¹, and to view Equation (12)'s Thinning Formalism as applying to any "Successor-based" axiom system α .

Equation (12)'s 2-part condition (and its various analogs) has been used to classify the generality of the Second Incompleteness Theorem. Let $\lceil \Psi \rceil$ denote Ψ 's Gödel number, and $\text{Prf}_\alpha^D(t, p)$ denote p is a proof of the theorem t from the axiom system α using the deduction method D . Then α will be said to recognize its **Cut-Localized D-consistency under** $\varphi(x)$ iff α can prove:

$$(15) \quad \forall p \{ \varphi(p) \rightarrow \neg \text{Prf}_\alpha^D(\lceil 0 = 1 \rceil, p) \}$$

The recent literature has sought to identify exactly what triples (φ, D, α) have the property that α can prove (15)'s self-reflecting statement about itself. Both negative and positive results, about the feasibility of constructing such triples, have been established.

A crucial negative result concerning the nature of Cut-Localized D-consistency was developed by Pudlák. The article [41] formalizes a generalization of the Second Incompleteness Theorem that shows that if D denotes Hilbert deduction and α represents

¹A "**SUCCESSOR-BASED ARITHMETIC**" axiom system α can be *thought of intuitively* as an axiomatic framework that recognizes Successor as a total function and which treats Addition, Multiplication and Exponentiation as three binary and ternary relations, denoted as $A(x, y, z)$, $M(x, y, z)$, and $E(x, y)$, rather than as formally total functions. By this, we mean that α will possess a finite set of Π_1 -like axioms built out of the atomic objects of a successor function symbol and the five predicate symbols for the equality, less-than, $A(x, y, z)$, $M(x, y, z)$, and $E(x, y)$ relations, such that the basic associative, commutative, distributive and identity axioms for addition and multiplication, as well as the inductive definition of the exponent operation, can be expressed by such Π_1 -like axioms. For instance, Equations (13) and (14) are examples of such Π_1 -like axioms, declaring that addition satisfies the commutative principle, and that the exponent operation satisfies its conventional inductive axiomatic definition.

$$(13) \quad \forall x \forall y \forall z \{ A(x, y, z) \longrightarrow A(y, x, z) \}$$

$$(14) \quad \forall x \forall y \forall z \{ [E(x, y) \wedge A(y, y, z)] \longleftrightarrow E(\text{Successor}(x), z) \}$$

IMPORTANT ADDED COMMENT about this Definition: There are several possible alternate somewhat broader definitions of a "*Successor-Based Arithmetic*" axiom system α , where both Equation (12)'s Thinning Formalism and Theorem *'s variation of the Second Incompleteness will continue to hold. For example, it follows from Benett's dissertation [6] that our preceding definition needs no explicit reference to $E(x, y)$'s exponent predicate (because it is Δ_0 definable in terms of addition and multiplication). Also, the opening chapters of Nelson's book [34] essentially implied that we may permissibly start with weaker principles than the associative, commutative and distributive axioms for addition and multiplication. However for the purposes of the current chapter's abbreviated literature survey, the preceding definition for "Successor-Based Arithmetics" is essentially adequate for understanding the main aspects of Equation (12)'s Thinning Formalism and its use by Theorem *.

any consistent extension of Q , then α must be unable to prove (15)’s statement about the “cut” φ .

The implications of Pudlák’s theorem can be further appreciated when one reviews some additional work by Nelson, Solovay and Wilkie-Paris. One of the opening chapters of Nelson’s book [34] explored an axiom system, called Q_0 , that had a conventional interpretation for the successor function, but which allowed for unconventional interpretations for addition and multiplication. By employing definable cuts as an intermediate concept, Nelson observed that this axiom system had many properties similar to the Tarski-Mostowski-Robinson axiom system Q [58]. Wilkie-Paris [64] used the theory of definable cuts to prove several new versions of the Second Incompleteness Theorem in [64] — one of which indicated that $I\Sigma_0 + Exp$ is unable to prove the Hilbert consistency of Q . Solovay [52] observed that one could combine the Pudlák theorem (from the prior paragraph) with the formalisms of Nelson and Wilkie-Paris to establish the following result:

Theorem * (Solovay’s generalization of Pudlák’s Theorem 2.3 from [41] using the additional mathematical methodologies of Nelson and Wilkie-Paris [34, 64], as approximately summarized in [52]’s 4-page proof.) *No consistent axiom system, satisfying the “Successor-Based Arithmetic Criteria” defined formally by Footnote 1, can possibly prove the non-existence of a Hilbert-proof of $0 = 1$ from itself.*

Aside from the work of Pudlák, Solovay and Wilkie-Paris already mentioned, there are many other generalizations of the Second Incompleteness Theorem for weak logics [1, 2, 3, 4, 5, 7, 8, 9, 11, 14, 15, 20, 21, 23, 29, 36, 37, 38, 40, 43, 45, 57, 66, 70]. The reason for our special interest in Theorem * is because this theorem, as described in reference [52], can help identify the precise circumstances where it is *infeasible* to develop an introspectively unified logic. In particular, [67, 72] have shown it is possible to devise introspective systems (α, D) when either:

A: D represents Semantic Tableaux deduction and α is an axiom system that recognizes addition but not multiplication as a total function [65, 67], or

B: D represents Hilbert deduction and α is a system that drops the axiom that addition is a total function and replaces it with the weaker “additive naming convention” of [72].

The Theorem * is relevant to the topics (A) and (B) because it shows that the natural hybrid of these two introspective logics is infeasible — for the case where D corresponds to Hilbert deduction and α recognizes addition (or even merely successor) as total.

Hence the contrast between Theorem *’s negative result with the positive results from Items (A) and (B) leads one to want to identify the exact juncture where the Second

Incompleteness Theorem takes force among axiom systems that recognize addition as a total function. More precisely, it raises the following question:

Which precise class of deduction methods D shall allow for an introspectively unified logic when α recognizes addition as a total function?

One part of an essentially 3-part answer to Question # was contained in our recent conference paper [70]. It showed that there existed a fixed Π_1^* theorem W , valid in the standard model of arithmetic (and actually provable from Peano Arithmetic), such that no consistent axiom system α can simultaneously prove W , recognize addition as a total function and verify its own self-consistency under either of the $\text{Tab}-\Sigma_2^*$ or $\text{Tab}-\Pi_2^*$ deduction methods. In effect, this means that the modus ponens rules for either Σ_2^* or Π_2^* sentences are sufficient to enact the power of the Second Incompleteness Theorem for essentially all axiom systems α which recognize addition as total. (The article [70] also explains how its results are related to some techniques that were developed by Adamowicz, Paris, Pudlák, Solovay, Takeuti, Wilkie, Zbierski and ourselves in [1, 4, 38, 41, 52, 57, 64, 68].)

Section 5 will present the second part of our 3-way answer to Question #. It will explore this question from an essentially opposite perspective as [70] by seeking a positive rather than negative-oriented answer to Question #. By analyzing the properties of Section 3's $\text{IS}_D(A)$ system, Section 5 will show that an introspectively unified logic (α, D) can simultaneously recognize its self-consistency under a modus ponens principle at the level of Σ_1^* and Π_1^* formulae and also retain an ability to prove all Peano Arithmetic's Π_1^* theorems.

A third part of our answer to the Question # will appear in Section 6. It will show that a close cousin of $\text{IS}_D(A)$, which Definition 3 had called $\text{NS}_D^{k,m}(A)$, is inconsistent. Hence, the combination of our positive and negative results will help characterize the threshold points where the Second Incompleteness Theorem takes force.

The prior literature has used an approach quite different from $\text{IS}_D(A)$'s Group-3 axiomatic mechanism for formalizing a type of respect in which an axiom system α can evade the Incompleteness Theorem and obtain a partial understanding of its own self-consistency. It has sought to accomplish this objective by formalizing some Definable Cuts φ such that an axiom system α has an ability to prove the validity of its own Cut-Localized D-consistency within the domain of φ 's definable cut (as is formally defined by Equation (15)). Some examples of this type of phenomena are listed below:

- A:** In a context where D denotes a second order generalization of Gentzen's sequent calculus and α is a formalization of arithmetic among the natural numbers, the Kreisel-Takeuti system [27, 56] can prove Equation (15)'s statement about its own D-consistency.
- B:** Nelson [34] demonstrated that a variant of the axiom system Q (with linear ordering) can corroborate Equation (15)'s statement about its own Cut-Localized

consistency when D denotes Herbrand deduction and the cut φ is carefully chosen.

- C:** Pudlák has formalized a major generalization of Items A and B that has very far-reaching and generic implications. The article [41] demonstrates that every “sequential” [20] axiom system α of finite cardinality can be associated with a definable cut φ such that α can prove Equation (15) is valid for φ when D denotes any cut-free method of deduction. (Indeed for an arbitrary definable cut Υ , this cut φ can be chosen to be a thinning of Υ .) At the same time, [41] showed that no consistent $\alpha \supset Q$ can prove any form of (15)’s sentence about itself when D denotes Hilbert deduction.
- D:** While GB Set Theory can prove its global Hilbert consistency is equivalent to the global Hilbert consistency of ZF Set Theory, Hájek, Švejdar and Vopěnka [55, 63] have shown GB views these theories differently inside the ranges of definable cuts. Thus, Vopěnka-Hájek [63] identified a definable cut φ where GB Set Theory can prove Equation (15)’s validity when D denotes Hilbert deduction and α denotes ZF Set Theory. Moreover, Švejdar [55] has generalized this phenomena with regards to interpretability. These results are surprising because Pudlák [41] subsequently proved GB can never verify its own Hilbert consistency on any of Equation (15)’s definable cuts. Thus, GB will view its Hilbert consistency as equivalent to ZF’s Hilbert consistency in a global sense — *but not in a sense localized* within each definable cut.
- E:** The notion of relative interpretability was made explicit as early as 1953 by Tarski-Mostowski-Robinson [58]. This topic and lattices of interpretability have been systematically studied by Feferman, Kreisel, Lindström, Montague, Mycielski, Orey, Pudlák and Švejdar [11, 12, 32, 31, 33, 35, 28, 54, 55, 39]. Characterizations of relative interpretability for reflexive axiom systems with infinite cardinality were obtained by Hájek [17, 18, 19] using the Orey Compactness Theorem, and for finite theories by Friedman and Pudlák independently [14, 41] — with Visser subsequently extending some of these results and introducing the notion of a feasible interpretation in [59, 62]. (Friedman never published his results of [15], which instead have been surveyed in articles by Pudlák, Smoryński and Visser [43, 50, 59, 62] and which were also largely independently reconstructed by Pudlák.) For essentially all finite theories S and T , the theory S has an interpretation in T if and only if $\text{IS}_0 + \text{Exp}$ can prove T ’s Herbrand consistency implies S ’s Herbrand consistency.
- F:** Let $\text{Con}^\varphi(\text{IS}_0)$ denote the particular variation of Equation (15) where $\alpha = \text{IS}_0$ and where D designates Hilbert-style deduction. Krajíček [26] proved that for any IS_0 cut Υ there exists a *thinner* IS_0 cut φ such that the theorem $\text{IS}_0 + \text{Con}^\varphi(\text{IS}_0) + \neg \text{Con}^\Upsilon(\text{IS}_0)$ is consistent. Visser [61] has generalized this construct to show that many consistent axiom systems, such as Q , ACA_0 and GB-Set Theory, have the property that *no finite consistent extension of themselves*

implies there exists Hilbert-style proofs of $0=1$ from themselves *simultaneously positioned in every one of their definable cuts*.

The underlying framework behind the $IS_D(A)$ system (of Section 3) is significantly different from the results summarized by Items A–F. The main reason $IS_D(A)$ is of interest is because it is able to internally recognize its own self-consistency in a global sense (rather than in a more narrow localized respect using Equation (15)’s Definable Cut formula φ). On the other hand, the price we pay for this added degree of generality is quite considerable — since $IS_D(A)$ treats multiplication as a 3-way relation — rather than as a total function.

A key point that needs to be appreciated is that this trade-off is essentially unavoidable. This point can be understood by taking a look at the recent literature on weak axiom systems. Aside from the work of Pudlák, Solovay and Wilkie-Paris already mentioned, other recent papers about incompleteness for weak axiom systems (listed chronologically) include: Paris-Dimitracopoulos’s observation [37] that one cannot universally define thinnings of definable cuts that are closed automatically under exponentiation, several generalizations of the Second Incompleteness Theorem for Bounded Arithmetic by Buss and Ignjatovic [8, 9], Takeuti’s observation [57] that one *can define* an exponentially-small and super-exponentially-small proof predicate that has some useful features for proving incompleteness results for cut-free deduction, Adamowicz-Zbierski’s observations [1, 4] that $I\Sigma_0 + \Omega_1$ is unable to verify its cut-free consistency, and some further generalizations of Adamowicz-Zbierski’s result for $I\Sigma_0$ by Willard [66, 68] and more recently Salehi [48]. Moreover, [66, 68] has also extended this formalism to show that for each cut-free deduction method D , there exists a particular well-defined Π_1^* sentence, called W_D , such that every consistent system $\alpha \supset W_D$ is unable to simultaneously recognize multiplication as total and to also prove a theorem affirming its Level(0-) consistency under D . This result is relevant to any effort to hybridize the theorems from the articles listed in Items A–F with $IS_D(A)$ because it shows that such hybridizations *cannot globalize Equation (15)’s notion* of a Cut-Localized D-consistency, *and simultaneously extend* $IS_D(A)$ to recognize multiplication as total.

Our result about multiplication [66, 68] is different from the incompleteness theorems of Bezboruah-Shepherdson, Pudlák, Solovay and Wilkie-Paris in [7, 41, 52, 64] by applying to a paradigm where D is a cut-free rather than a cut-permissive deduction method. It establishes that any effort to find a cut-free deduction method D and an axiom system α — where α can prove its Level(0-) consistency under D without relying upon Equation (15)’s Definable Cuts — is fruitless when α recognizes multiplication as total and $\alpha \supset W_D$. Its incompleteness result explains why our goal will be to *drop the Π_2^* axiom that declares multiplication is total*, so that Section 5’s $IS_D(A)$ axiom system can obtain a Tab- U_1^* -List (or slightly greater) understanding of its own self-consistency.

This alternate perspective is useful partly because it is interesting to consider axiomatic formalisms that do not rely upon Equation (15)’s notion of a locally-defined

consistency property for gaining a partial understanding of their own self-consistencies. This approach clearly has its limitations because $IS_D(A)$ drops the important axiom that multiplication is total. However in a context where the philosophical paradox associated with Gödel's Second Incompleteness Theorem is inherently perplexing, it is of theoretic interest to consider $IS_D(A)$'s alternate properties.

A second reason for interest in $IS_D(A)$ is because a multiplicity of generalizations of the Second Incompleteness Theorem, including one especially relevant new effect delineated in Section 6, collectively imply that further efforts to improve $IS_D(A)$ will meet inherent limitations.

§5. The Consistency Preservation Property of $IS_D(\bullet)$. This chapter will be divided into three subsections. Section 5.1 will describe a general framework for determining which deduction methods D are associated with " $IS_D(\bullet)$ mappings" that are consistency-preserving. Section 5.2 and 5.3 will prove that these conditions are met respectively when D represents semantic tableaux and $Tab-U_1^*$ -List deduction.

Some readers may be feeling a bit puzzled by Section 3's announcement that a close cousin of $IS_D(\bullet)$, called $NS_D^{k,m}(\bullet)$, always fails to satisfy the consistency-preserving property. A 3-paragraph summary about the intuitive differences between $IS_D(\bullet)$ and $NS_D^{k,m}(\bullet)$ is provided by Section 6.1 of the next chapter.

It is unnecessary for the reader to examine Section 6.1's short 3-paragraph passage before he enters the current chapter. However, if a reader takes a quick glance at Section 6.1 at this juncture, he may find some of his most puzzling questions are answered.

5.1. Structural Aspects of Main Proof. The Appendix A describes one canonical method for assigning Gödel encodings to proofs using the $IS_D(A)$ axiom system. This encoding method is roughly the same as what Wilkie-Paris [64] had called a *natural B-adic encoding* — or a similar counterpart in the Hájek-Pudlák textbook [20]. Such B-adic encodings are maximally compressed in that their encoding for a proof p will have a bit-length approximately proportional to the effort for writing down such a proof by hand. The Appendix A's formalized description of our encoding conventions is provided only for the sake of completeness. Its reading can be easily omitted.

Some added notation is needed to help introduce this section's main theorem. Let us say a Π_1^* sentence $\forall x_1 \forall x_2 \dots \forall x_n \psi(x_1, x_2 \dots x_n)$ satisfies the $App\forall(a)$ condition when Equation (16) is true:

$$(16) \quad \forall x_1 \leq a \forall x_2 \leq a \dots \forall x_n \leq a \psi(x_1, x_2 \dots x_n)$$

Similarly, a Σ_1^* sentence $\exists x_1 \exists x_2 \dots \exists x_n \psi(x_1, x_2 \dots x_n)$ is said to satisfy $App\exists(b)$ when (17) is true.

$$(17) \quad \exists x_1 \leq b \exists x_2 \leq b \dots \exists x_n \leq b \psi(x_1, x_2 \dots x_n)$$

These definitions refer only to the ranges of a sentence's *unbounded* universal and existential quantifiers. Bounded quantifiers do not have their ranges changed similarly under these definitions.

DEFINITION 4. For any ordered pair (a, b) with $a > b \geq 1$, let us say an axiom system α satisfies the **Normed(a,b)** condition iff:

1. All α 's axioms are either Π_1^* sentences satisfying the $App\forall(a)$ condition or Σ_1^* sentences satisfying the $App\exists(b)$ condition.
2. α uses a language that employs the eight U-Grounding function symbols and absolutely no other function symbols.

(The second condition is designed to preclude α from employing a multiplication function symbol — or any of its many functional equivalents.)

Definition 5 will be the main engine used to prove $IS_D(\bullet)$'s consistency preservation property, for actually several different deduction methods D . It did not exist in our prior papers. It is helpful for both shortening several of our prior proofs — and for producing substantially stronger results:

DEFINITION 5. Let α denote an $\text{Normed}(a, b)$ system with $a > b \geq 1$, Φ denote some Π_1^* or Σ_1^* sentence, and p denote the Gödel number of some proof from α of the theorem Φ (using the deduction method D). Let $\theta < \frac{1}{3}$ be a positive constant, and p satisfy Equation (18).

$$(18) \quad p \leq \lceil \frac{a}{b} \rceil + 1$$

Then the deduction method D will be called θ -**Compactified** iff every such tuple (α, Φ, p, a, b) satisfies the following two conditions:

- A:** If Φ is a Σ_1^* theorem (whose proof p satisfies the above conditions), then Φ has an $App\exists(b \cdot p^\theta)$ level of validity.
- B:** Likewise if Φ is a Π_1^* theorem proved by p , it must have an $App\forall(\frac{a}{p^\theta})$ level of validity.

THEOREM 1. For some positive constant $\theta < \frac{1}{3}$, let us assume that D is a θ -Compactified deduction method. Then the $IS_D(\bullet)$ mapping transformation is a “consistency-preserving” operation. (Consistency Preservation was defined by Definition 1. It indicates that $IS_D(A)$ is automatically a consistent axiom system whenever A 's Π_1^* theorems are valid in the standard model.)

PROOF. Our justification of Theorem 1 will be a proof by contradiction. It will therefore begin by assuming that all A 's Π_1^* theorems are valid in the standard model but $IS_D(A)$ is inconsistent. It will show this assumption leads to a contradiction.

In our discussion, j_1 will represent a proof of some Π_1^* theorem, called say Φ , and j_2 will be a proof of the Σ_1^* theorem that is Φ 's negation. Also, m will denote the quantity $\text{Max}(j_1, j_2)$. If $IS_D(A)$ is inconsistent, then such a triple (j_1, j_2, m) will clearly exist. Let us therefore select such a triple whose third component has the minimum possible value for m .

It is obvious that all $IS_D(A)$'s Group-0 and 1 axioms are valid under the standard model. From our starting assumption that all A 's Π_1^* theorems are valid in the standard model, it immediately follows that all $IS_D(A)$'s Group-2 axioms are also valid in the standard model. Moreover, all these axioms can be encoded as Π_1^* sentences (when they are written in Prenex* form). Hence since $IS_D(A)$'s Group-3 axiom sentence becomes invalid only for integers greater than $m - 1$, $IS_D(A)$ is *certainly* a Normed($m - 1, 1$) system.

We already observed that the two Gödel numbers, j_1 and j_2 , that prove Φ and its negation, are integers no larger than m . These facts, Definition 5 and $IS_D(A)$'s Normed($m - 1, 1$) property immediately imply that that Φ has an $App\forall(\frac{m-1}{m^\theta})$ or greater level of validity and its negation has at least a $App\exists(m^\theta)$ level of validity.

These observations are all that is needed to bring our proof-by-contradiction to its desired conclusion. This is because if $g > h$, then it is clearly impossible for both a Π_1^* sentence Φ to have an $App\forall(g)$ level of validity and *for simultaneously* the Σ_1^* sentence corresponding to its negation to have an $App\exists(h)$ level of validity. *Yet precisely this pair of inherently incompatible conditions* is established, with $g = \frac{m-1}{m^\theta}$ and $h = m^\theta$ by the last sentence of the preceding paragraph (since $\theta < \frac{1}{3}$ implies $g > h$). Hence $IS_D(A)$ must be forced to be consistent, since the alternative hypothesis would otherwise force two inherently incompatible conditions to become simultaneously valid. $\dashv$

REMARK 1. The next two sections will apply Theorem 1 to prove that $IS_D(\bullet)$'s mapping transformation is consistency-preserving in the two particular cases where D represents semantic tableaux and Tab- U_1^* -List deduction. Although we will not have the page space to do so, there are many other examples where Theorem 1's formalism is also useful. For instance, it can be used to prove the $IS_D(\bullet)$ transformation is consistency-preserving in the additional cases where D represents the cut-free version of the sequent calculus, Herbrand deduction, Tab- Q_1^* -List deduction, or a variant of Hilbert deduction where each intermediate formula in a proof is required to have a Q_1^* format. In each of these cases, the proof of consistency preservation has a similar structure: It involves first doing a detailed case analysis showing the relevant method

D satisfies Definition 5's θ –Compactification condition and then applying Theorem 1's machinery.

REMARK 2. Proofs by contradiction are often quite difficult to visualize in mathematics because of their non-constructive nature. The core proof of Theorem 4.3 in [67] had this problem. On the other hand, our current proof of Theorem 1 is much easier to visualize. Thus in addition to strengthening [67]'s results, Theorem 1's machinery and its formal applications in the next two sections are also much more easy to visualize.

REMARK 3. Section 4's literature survey had indicated that Solovay had combined the formalisms of Nelson, Pudlák and Wilkie-Paris to prove that no axiom system recognizing Successor as a total function can verify its own Hilbert consistency. Also, our literature review had mentioned that [70] has generalized this prior result to establish that there exists essentially no introspectively unified logics (α, D) where α can recognize addition as a total function and D has the power of either the Tab- Π_2^* or Tab- Σ_2^* –List deductive methodologies. At this juncture, it is possible to explain intuitively why these generalizations of the Incompleteness Theorem hold. It is because their deductive methods do not satisfy Parts (A) and (B) of Definition 5's requirements for θ –Compactification. Hence, some significant generalizations of the Second Incompleteness Theorem take place at exactly the level where Definition 5's formalism becomes no longer applicable.

REMARK 4. It also noteworthy that Parts (A) and (B) of Definition 5's requirements are impossible to satisfy when Part-2 of the definition of a Normed(a,b) system is expanded to allow for multiplication to appear as a ninth function symbol. This is intuitively because multiplication has a faster growth property than addition, which will overwhelm Definition 5's constraints. Moreover, a Level(0-) styled semantic tableaux generalization of the Second Incompleteness Theorem in [68] implies that no useful analog of Definition 5 can be found for axiom systems recognizing multiplication as a total function, *under any possible deduction method* D , whether cut-free or otherwise.

5.2. Formal Analysis of Semantic Tableaux Deduction. This section will prove semantic tableaux satisfies Definition 5's θ –Compactification property when $\theta = \frac{1}{4}$. In conjunction with Theorem 1 and with some further results proven in Section 5.3, this result will imply that the $IS_D(\bullet)$ mapping is a consistency-preserving transformation when D designates either semantic tableaux or Tab- U_1^* –List deduction.

Notation: Given an axiom system Z , a **Z–Based Deduction Tree** is defined as a tree of sentences — where every node of this tree is either a formal axiom of Z or a deduction from an axiom, using one of Section 2's eight rules for Tableaux-style deductions. This definition differs from a Semantic Tableaux proof in two respects. These are:

- i: Since a Z -Based Deduction Tree is not technically “a proof”, its root will store an axiom of Z (rather than store the negation of a target theorem).
- ii: Unlike Section 2’s definition of a “Semantic Tableaux proof tree”, a Z -Based Deduction Tree will not be required to have every one of its branches “closed” by a pair of contradictory sentences. (Indeed, if Z is consistent, then its Z -Based Deduction Trees will always contain at least one branch that is not “closed”.)

Also, the notation (below) is used when discussing Z -Based Deduction Trees:

1. The **U-Length** of a node sentence s is defined to be the number of U-grounding function symbols appearing in this sentence.
2. The **U-Depth** of a node sentence s , henceforth denoted as $\Delta(s)$, is defined to be the sum of s ’s U-length with the U-lengths of all its ancestors.
3. The **U-Height** of a Deduction Tree is the maximum U-Depth value among its leaves.

Some notation is needed to start our discussion. For any ordered pair (a, b) with $a > b \geq 1$, let Z denote an Normed(a, b) system, T denote a Z -Based Deduction Tree, and β denote a branch of T . Let $\text{VAL}(\bullet)$ denote a “valuation” function that maps each parameter symbol u (appearing in the branch β) onto an integer $\text{VAL}(u)$, satisfying the following two constraints:

- I: Suppose a node s of the branch β introduces a new parameter symbol u via one of Section 2’s $\exists$ quantifier Elimination rules. Then Item 2’s $\Delta(s)$ symbol will satisfy the inequality:

$$(19) \quad \text{Val}(u) \leq b \cdot 2^{\Delta(s)}$$

- II: Every node-sentence s along the branch β will satisfy the following 3-part condition:

II-a: If s is a Π_1^* sentence then it will possess an $\text{App}\forall(a)$ level of validity under the valuation VAL .

II-b: If s is a Σ_1^* sentence then it will be $\text{App}\exists(b)$ valid under VAL .

II-c: If s is a Δ_0^* sentence, then it will be a logically valid statement under the valuation’s scheme for assigning an integer value $\text{VAL}(u)$ to each parameter symbol u . (This third requirement can be thought of as the intersection of requirements II-a and II-b because their combination corresponds to II-c’s requirement when a sentence contains absolutely no unbounded quantifiers. This is because our definitions of $\text{App}\forall(c)$ and $\text{App}\exists(c)$ involves changing a sentence’s unbounded quantifiers so that their ranges are bound by c , but it implies no similar change in a *bounded* quantifier’s range.)

Henceforth, a node N in the deduction tree T will be called **Positive(a,b)** iff the union of N ’s sentence with its ancestors in the deduction tree T satisfy the conditions I and

II under some valuation VAL. Also, a root-to-leaf branch β will be called **Positive(a,b)** when there exists a valuation VAL where all its nodes satisfy the conditions I and II.

LEMMA 1. *For any $a > b \geq 1$, let Z denote an Normed(a, b) system and T denote a Z -Based Deduction Tree whose U-Height is less than $\text{Log}_2 a - \text{Log}_2 b$. Then T possesses at least one Positive(a, b) branch.*

PROOF. The easiest manner to prove Lemma 1 is to first give an algorithm that constructs T 's Positive Branch and then to prove the correctness of this algorithm. Our algorithm for constructing this branch β and its valuation VAL from the input (a, b, T) will be called $\text{Probe}(a, b, T)$. Its four steps are listed below:

1. The top node along the path β will always be T 's root.
2. Suppose the first i nodes along the path β are $N_1, N_2, \dots, N_i$ and N_i has two children, denoted as N_e and N_f . In this case, the deduction tree T has used either the $\vee$ or $\rightarrow$ Elimination rule to justify this binary split, and we will let Ψ_e and Ψ_f denote the two sentences stored in these two nodes. The algorithm $\text{Probe}(a, b, T)$ will make the "left child" N_e constitute the next element along β 's path when the valuation VAL has Ψ_e satisfy condition II. Otherwise, it will make N_f be β 's next node.
3. If the first i nodes in β 's path are $N_1, N_2, \dots, N_i$ and N_i has only one child, denoted as N_{i+1} , then the algorithm $\text{Probe}(a, b, T)$ will "attempt" to make N_{i+1} into β 's next node. If N_{i+1} introduces a new parameter symbol u , this "attempted step" will also assign $\text{VAL}(u)$ the smallest integer value satisfying the Positive(a, b) condition (consistent with earlier assignments the valuation VAL had already made). The proof that this "attempted extension" of β will always successfully satisfy the Positive(a, b) condition has nine subcases, labeled (a) through (i), appearing on the next several pages.
4. The procedure $\text{Probe}(a, b, T)$ will iteratively repeat Steps 2 and 3 to make the path β become longer and longer until it reaches the tree's leaf-level.

The remainder of our proof of Lemma 1 will be devoted to establishing that the procedure $\text{Probe}(a, b, T)$ will successfully construct a Positive(a, b) branch when the deduction tree T satisfies this lemma's hypothesis. The Principle of Induction will be employed to justify this claim. Our inductive proof shall assume that the first i nodes $N_1, N_2, \dots, N_i$ along the path β satisfy the positive condition, and it will use this fact to deduce N_{i+1} is also positive. Our inductive proof will be divided into nine sub-cases because we must separately consider the possibilities that N_{i+1} is an axiom of Z or is deduced via one of the eight Tableaux-style Elimination rules, that were formally defined in Section 2.

a. The Case where N_{i+1} stores one of Z 's axioms. Any axiom trivially satisfies Part-I of the definition of Positiveness (simply because it contains no parameter symbols u). Moreover by definition, each axiom of an Normed(a, b) system automatically satisfies Requirement II. $\square$

b. The Case where N_{i+1} is generated by the “Unbounded Version” of the $\exists$ -Elimination Rule (i.e. Section 2's Deduction Rule 5): This Elimination rule allows N_{i+1} to represent a sentence $\psi(u^*)$, containing a new parameter symbol u^* , when an ancestor N_j of N_{i+1} represents the sentence $\exists v \psi(v)$. Since N_j can be inductively presumed to satisfy Part II-b of the definition of Positiveness, it is immediately apparent that u^* can be assigned a valuation number $\text{VAL}(u^*)$ enabling N_{i+1} to also satisfy the Positive(a, b) condition. $\square$

c. The Case where N_{i+1} is generated by the “Bounded” type $\exists$ Elimination Rule (i.e. Deduction Rule 6): This case is analogous to Item b, although it is more complicated. In this case, an ancestor N_j of N_{i+1} will represent the formal sentence $\exists v \leq t \psi(v)$, and N_{i+1} is a deduction from N_j introducing a new symbol u^* and storing the sentence $u^* \leq t \wedge \psi(u^*)$. Let Δ denote N_{i+1} 's U-Depth, and let L denote N_{i+1} 's U-Length. Since N_j can be inductively presumed to satisfy Part I of the definition of Positiveness and because N_j has a U-depth $\leq \Delta - L$, it follows that every parameter symbol $u_1, u_2, \dots, u_m$ appearing in N_j will satisfy $\text{VAL}(u_i) \leq b \cdot 2^{\Delta-L}$. In turn, this implies that $\text{VAL}(t)$ has an integer value no greater than $b \cdot 2^\Delta$ (because t is constructed by taking input values no larger than $b \cdot 2^{\Delta-L}$ and applying no more than L iterations of growth-functions using these inputs — where the Double operation is the fastest available growth function). Since N_j can be inductively assumed to satisfy Part II-c of the definition of Positiveness and it is a sentence of the generic form $\exists v \leq t \psi(v)$, we can certainly construct a valuation where $\text{VAL}(u^*) \leq b \cdot 2^\Delta$ and where $\text{VAL}(u^*)$ assures that N_{i+1} will consequently satisfy Parts I and IIc of the Positive(a, b) condition. $\square$

d. The Case where N_{i+1} is generated by the “Unbounded Version” of the $\forall$ Elimination Rule (i.e. Deduction Rule 7): This Elimination rule allows N_{i+1} to represent a sentence $\phi(t)$, when an ancestor N_j of N_{i+1} stores the sentence $\forall v \phi(v)$ and t represents a term in the U-Grounding language. Our Deduction Rule 7 further requires that each parameter symbol u , appearing in Rule 7's term t , must be previously introduced by an ancestor of N_{i+1} , employing the $\exists$ Elimination Rule.

It is automatic from the last fact that N_{i+1} will satisfy Part-I of the definition of the Positiveness (simply because there are no new parameter symbols u introduced by the $\forall$ -Elimination Rule). To show that N_{i+1} also satisfies the Part-II requirement, Δ and L will again denote N_{i+1} 's U-Depth and U-Length. Then since t has no more than L function symbols and no newly introduced parameter symbols, we can apply Case(c)'s methodology to again conclude that $\text{VAL}(t) \leq b \cdot 2^\Delta$ (see footnote ²).

²The same argument, as before, applies because each parameter symbol u inside t can again be inductively presumed to satisfy $\text{VAL}(u) \leq b \cdot 2^{\Delta-L}$. This in turn implies $\text{VAL}(t) \leq b \cdot 2^\Delta$ because t

Hence since T has a U-Height less than $\text{Log}_2 a - \text{Log}_2 b$, we get $\text{VAL}(t) \leq a$. This last inequality, combined with the inductive assumption that N_j satisfies Part II-a of $\text{Positive}(a, b)$ condition, immediately implies N_{i+1} also satisfies Part-II of the definition of $\text{Positive}(a, b)$. $\square$

e. The Case where N_{i+1} is generated by the “Bounded Version” of the $\forall$ Elimination Rule (i.e. Deduction Rule 8): This case can be viewed as a degenerate version of the Case (d) above, and it thus requires no proof.

f. The Case where N_{i+1} is generated by the $\wedge$ Elimination Rule: Trivial because a sentence Υ will automatically satisfy the $\text{Positive}(a, b)$ Condition when some ancestor of it storing the sentence $\Upsilon \wedge \Gamma$ does. $\square$

g. The Case where N_{i+1} is generated by a $\neg$ Elimination Rule: It is, once again, trivial that a sentence Υ will automatically satisfy the $\text{Positive}(a, b)$ Condition when some ancestor of it storing the sentence $\neg \neg \Upsilon$ does. Also, a similar easy argument applies to the other variants of the $\neg$ Elimination (formally defined by Section 2’s Deduction Rule 2). $\square$

h. The Case where N_{i+1} is generated by the $\vee$ Elimination Rule: This rule introduces a pair of sibling nodes Υ and Γ when they have a common ancestor $\Upsilon \vee \Gamma$. Since the inductive hypothesis implies $\Upsilon \vee \Gamma$ satisfies the $\text{Positive}(a, b)$ Condition, one of Υ or Γ must also satisfy this condition. Our algorithm Probe will automatically select this satisfying node. $\square$

i. The Case where N_{i+1} is generated by the $\rightarrow$ Elimination Rule: Essentially the same as the preceding Case h. $\square$ $\dashv$

LEMMA 2. *Again suppose $a > b \geq 1$, that Z is a $\text{Normed}(a, b)$ axiom system, and that T denotes a Z -Based Deduction Tree. Then no $\text{Positive}(a, b)$ Branch of Z ’s deduction Tree T can be “closed” by containing simultaneously some sentence Υ and its negation.*

PROOF. Our proof of Lemma 2 will be a proof-by-contradiction. We will thus entertain the hypothesis that T has a closed $\text{Positive}(a, b)$ branch containing simultaneously some sentence Υ and its negation, and show that this assumption leads to a contradiction.

From the fact that all axioms in a $\text{Normed}(a, b)$ system are Prenex* sentences of the Π_1^* and Σ_1^* types, it follows that a branch can contain a sentence Υ and its *strictly encoded* negation of the *exact form* “ $\neg \Upsilon$ ” only when both Υ and $\neg \Upsilon$ are Δ_0^* sentences (i.e. see footnote ³).

applies no more than L iterations of the doubling and addition functions to inputs whose initial sizes are bounded by $b \cdot 2^{\Delta-L}$.

³The classes of Δ_0^* formulae were defined in Section 2. They consisted of any sentence all of whose quantifiers were bounded. On the other hand, our defining conventions implied that if a sentence Υ was Σ_1^* or Π_1^* **BUT NOT** Δ_0^* , then it must contain one or more unbounded quantifier to the left of all other logical symbols, including the “ $\neg$ ” symbol. This means that if Υ is either a proper Σ_1^* or Π_1^* sentence,

However, Part II-c of the definition of Positiveness indicates each Δ_0^* sentence, appearing in a $\text{Positive}(a, b)$ branch, must be a valid statement under the branch's valuation function. Hence, the concerned valuation must cause Υ and $\neg \Upsilon$ to become simultaneously valid.

The latter property of Υ and $\neg \Upsilon$ is, of course, inherently infeasible. Hence, our proof-by-contradiction has reached its goal because it has shown that it is infeasible for T to possess a closed $\text{Positive}(a, b)$ branch. $\neg$

We will now prove our second theorem about the θ –Compactification property.

THEOREM 2. . *Semantic tableaux satisfies the θ –Compactification property when $\theta = \frac{1}{4}$.*

PROOF. Lemmas 1 and 2 will enable us to show that semantic tableaux satisfies Parts (A) and (B) of the definition of θ –Compactification:

Demonstration that Semantic Tableaux Deduction satisfies Part (A) of the definition for θ –Compactification: Let us suppose — for the sake of establishing a contradiction — that a Σ_1^* sentence Φ has a proof $p \leq \lceil \frac{a}{b} \rceil + 1$ from a $\text{Normed}(a, b)$ system α while Φ 's $\text{App}\exists(p^\theta \cdot b)$ interpretation is simultaneously fallacious. Then p 's root will represent the negation of Φ rewritten in Prenex* form, and this Π_1^* representation of $\neg \Phi$ will be have an $\text{App}\forall(p^\theta \cdot b)$ validity level.

Let Z denote the axiom system that is the union of this Π_1^* representation for $\neg \Phi$ with α . It is easy to verify that Z is a $\text{Normed}(p^\theta \cdot b, b)$ system (see footnote ⁴).

Our next observation is rather technical. The Appendix A's method for encoding proof-trees, as well as any other reasonable encoding methodology, will clearly satisfy Equation (20)'s inequality.

$$(20) \quad \text{U-Height}(p) < \frac{1}{5} \text{Log}_2(p)$$

Since Theorem 2's hypothesis indicated that $\theta = \frac{1}{4}$ and Z was a $\text{Normed}(p^\theta \cdot b, b)$ system, Equation (20) implies that the pair (p, Z) must consequently satisfy Lemma

then technically $\neg \Upsilon$ is neither Σ_1^* nor Π_1^* because a negation symbol appears to the left of its unbounded quantifiers. Hence since no sentence along our explored branch β has a complexity higher than Σ_1^* or Π_1^* and because *both* Υ and $\neg \Upsilon$ appear in this $\text{Positive}(a, b)$ branch, the forced conclusion is that Υ is a Δ_0^* sentence.

⁴The opening sentence of our proof assumed that $p \leq \lceil \frac{a}{b} \rceil + 1$. Also, Theorem 2's hypothesis indicates $\theta = \frac{1}{4}$. These identities imply $p^\theta \cdot b \leq a$. Since the opening sentence of our proof presumed α was a $\text{Normed}(a, b)$ system, this $p^\theta \cdot b \leq a$ inequality implies α is also a $\text{Normed}(p^\theta \cdot b, b)$ system. Since $\neg \Phi$ possesses an $\text{App}\forall(p^\theta \cdot b)$ level of validity when it is written as a Π_1^* sentence, the union of $\neg \Phi$ with α must thus also be a $\text{Normed}(p^\theta \cdot b, b)$ system.

1's hypothesis (i.e. see footnote ⁵). Hence, Lemma 1 implies that p must have a positive branch.

From Lemma 2, this implies that such a branch of p will fail to be closed. Hence, p cannot be a formal semantic tableaux proof (because at least one of its branches fails to be "closed"). The latter observation brings our proof-by-contradiction to its desired end by contradicting the initial assumption that p was a proof of Φ . $\square$

Demonstration that Semantic Tableaux Deduction satisfies Part (B) of the definition for θ —Compactification: This proof appears in Appendix B. Briefly summarized, it is roughly a permutation of the ideas from Part (A)'s proof — with the roles of the universal and existential quantifiers now reversed. (As with Part (A)'s proof, the main idea behind (B)'s proof rests on using the mathematical machineries of Lemmas 1 and 2 as the main underlying interim steps.) $\dashv$

THEOREM 3. *The $IS_D(\bullet)$ mapping is a consistency-preserving transformation when D denotes the semantic tableaux. (In other words, $IS_D(A)$ is automatically consistent when all A 's Π_1^* theorems are valid in the standard model of the natural numbers.)*

PROOF. An immediate consequence of Theorems 1 and 2 because the latter result shows $IS_D(\bullet)$ satisfies the Theorem 1's requirements for consistency preservation. $\dashv$

REMARK 5. (concerning the formal nature of Theorems 2 and 3): The meticulous reader may have noticed that the proofs in this section relied upon some very exacting definitions for semantic tableaux deduction and for Π_1^* and Σ_1^* sentences. These notation conventions were that:

- I:** The root in Φ 's semantic tableaux proof was required to consist of the sentence $\neg \Phi$ rewritten in Prenex* form. (This requirement was announced during Section 2's formal definition of a semantic tableaux proof. Some but not all of the prior literature has used a similar definition of a semantic tableaux that includes a "Prenex Normalized" requirement for representing $\neg \Phi$.)
- II:** If Υ is a proper Σ_1^* formula then the *formalized sentence* " $\neg \Upsilon$ " was technically viewed in our discussion as *not constituting* a formal Π_1^* sentence (because a $\neg$ symbol had appeared to the left of its unbounded $\exists$ -quantifiers.) However, this *formalized sentence* " $\neg \Upsilon$ " is clearly logically equivalent to a Π_1^* sentence. This type of strict definition for the Π_1^* and Σ_1^* classes helped simplify the discussion in Footnote 3 of Lemma 2's proof.

The notation conventions in I and II were used because they have simplified the structure for the proofs of Lemma 2 and Theorem 2. They also obviously raise the following natural question:

⁵To verify that (p, Z) satisfies Lemma 1's hypothesis, it is sufficient to ascertain that p 's U-Height satisfies the inequality of $U\text{-Height}(p) < \log_2(p^\theta \cdot b) - \log_2 b = \theta \cdot \log_2 p$. The combination of Theorem 2's assumption that $\theta = \frac{1}{4}$ and Equation (20) immediately implies the validity of the claimed $U\text{-Height}(p) < \theta \cdot \log_2(p)$ inequality.

* Would the results in Theorems 2 and 3 remain valid if Item I and II's particular notation conventions were dropped? After all, one would certainly prefer these two propositions to be *sufficiently robust* so that they do not rely upon such exacting assumptions?

The answer to * is that the conventions used by I and II are indeed *completely unnecessary*. However, the formal proofs of Lemma 2 and Theorem 2 *become significantly simplified* when the conventions I and II are employed. In particular, the Footnote 3 in Lemma 2's proof and the opening paragraphs in the Cases (A) and (B) of Theorem 2's proof gain pleasantly simpler structures when the conventions from Items I and II are followed.

5.3. Formal Analysis of Tab- U_1^* -List Deduction. The $IS_D(A)$ axiom system of Theorem 3 — with or without Remark 5's added considerations — is already much stronger than the $IS(A)$ system of [67] because it has a capacity to recognize its semantic tableaux self-consistency in a Level(1) rather than in a Level(0-) formalized sense. The natural follow-up question is whether Theorem 3's results can be possibly improved yet further, beyond Remark 5's proposed omitting of the Requirements I and II ?

On the positive side of the ledger, our research has found that Theorem 3 can be strengthened by having D represent either Tab- U_1^* or Tab- Q_1^* -List deduction. These strengthened versions of Theorem 3 are close to maximal effects, since the conventional force of the Second Incompleteness Theorem takes place at the levels of both Tab- Σ_2^* or Tab- Π_2^* -List deduction [70].

For both Tab- U_1^* or Tab- Q_1^* -List deduction, the proofs of their generalizations of Theorem 3 rests on a 2-step process that first shows these deduction methods satisfy the θ -Compactification property and then uses Theorem 1 to finish the analysis. Our discussion will focus on Tab- U_1^* -List deduction because its formal analysis is pleasantly shorter.

THEOREM 4. *Tab- U_1^* -List deduction satisfies the θ -Compactification property when $\theta = \frac{1}{4}$.*

PROOF. We will use the Principle of Induction to infer Theorem 4 from Theorem 2. Let P denote the Gödel number of a sequence of pairs, $(t_1, p_1), (t_2, p_2), \dots (t_n, p_n)$, which constitute a Tab- U_1^* -List proof of a theorem Φ from a Normed(a, b) axiom system called α . The formalism for θ -Compactification (given in Definition 5) indicates that this construct applies to any proof P which satisfies (21)'s inequality.

$$(21) \quad P \leq \lceil \frac{a}{b} \rceil + 1$$

Let Q denote the quantity defined below:

$$(22) \quad Q = \lfloor \frac{P}{p_n} \rfloor$$

A later step in our proof will use the fact that Equations (21) and (22) imply (23)'s validity.

$$(23) \quad p_n \leq \lceil \frac{a}{b \cdot Q} + 1 \rceil$$

Our inductive construction is allowed to assume that each of the intermediate theorems of $t_1, t_2, \dots, t_{n-1}$ have $\text{Tab}-U_1^*$ -List proofs no larger than Q (because this invariant holds under all typical encoding conventions, including the particular encoding method delineated in Appendix A). In this context, our proof-by-induction may presume that for each integer $i \leq n-1$, the sentence t_i will have

- A:** an $\text{App}\exists(Q^\theta \cdot b)$ level of validity when t_i is a Σ_1^* sentence, and
- B:** an $\text{App}\forall(\frac{a}{Q^\theta})$ level of validity when t_i is Π_1^* .

The union of α with the sentences $t_1, t_2, \dots, t_{n-1}$ is thus an $\text{Normed}(\frac{a}{Q^\theta}, Q^\theta \cdot b)$ axiom system. Let us call this axiom system α^* .

We will next use the fact that p_n is the Gödel number of a semantic tableaux proof from the axiom system α^* of the theorem Φ . This fact combined with Theorem 2, Equation (23) and α^* 's $\text{Normed}(\frac{a}{Q^\theta}, Q^\theta \cdot b)$ property implies (via footnote ⁶) that:

- i:** Φ has an $\text{App}\exists((Q \cdot p_n)^\theta \cdot b)$ level of validity when it is a Σ_1^* theorem,
- ii:** Φ has an $\text{App}\forall(\frac{a}{(Q \cdot p_n)^\theta})$ level of validity when it is a Π_1^* theorem.

Since Equation (22) indicates $P \geq Q \cdot p_n$, the invariants (i) and (ii) in turn imply:

- A:** Φ has an $\text{App}\exists(P^\theta \cdot b)$ level of validity when it is a Σ_1^* theorem,
- B:** Φ has an $\text{App}\forall(\frac{a}{P^\theta})$ level of validity when it is a Π_1^* theorem,

The latter two observations show that $\text{Tab}-U_1^*$ -List deduction is θ -Compactified. $\dashv$

THEOREM 5. *The $\text{IS}_D(\bullet)$ mapping is a consistency-preserving transformation when D denotes the $\text{Tab}-U_1^*$ deduction method. (In particular, this means that $\text{IS}_D(A)$ is automatically consistent when all A 's Π_1^* theorems are valid under the standard model.)*

⁶The second paragraph in Theorem 4's proof indicated that α^* was a $\text{Normed}(\frac{a}{Q^\theta}, Q^\theta \cdot b)$ axiom system. Also, Theorem 2 indicated that semantic tableaux deduction satisfied the θ -Compactification property. From the formal description of this property (in Definition 5), it is apparent that every semantic tableaux proof p_n of a theorem Φ (from the axiom system α^*) shall automatically satisfy the invariants (i) and (ii) when p_n satisfies the inequality of:

$$p_n \leq \lceil \frac{a}{b \cdot Q^{2\theta}} \rceil + 1$$

The final point is that Equation (23) and the fact that $\theta = 1/4$ imply that p_n does, in fact, satisfy the above inequality. Hence, the above construction shows that invariants (i) and (ii) do hold. $\square$

PROOF. An immediate consequence of Theorems 1 and 4 because the latter result shows that $IS_D(\bullet)$ satisfies the Theorem 1's requirements for consistency preservation. $\dashv$

REMARK 6. A class of self-justifying statements, called the Tangibility Reflection Principles, was defined in [67]. It is possible to hybridize our current results with [67]'s formalism — so that $Tab-U_1^*$ deduction can be incorporated into axioms systems verifying their tangibility principles. (These hybrids are also compatible with the $Tab-Q_1^*$ and the other 1-deductive methods of Remark 1.) The details for constructing such hybrids are very lengthy. They are therefore not done here. Our article [67] explains why the tangibility reflection principle is actually very useful.

REMARK 7. This paper was submitted to the *Journal of Symbolic Logic* in August of 2004, and we have not made any major changes in the manuscript since then except essentially for responding to the referee's suggestions, all of which were very helpful. During the 9-month period following the submission of this manuscript, we continued working on this on-going research project, and found an interesting application of Theorem 5's formalism. This application requires a rather lengthy description, and its characteristics are summarized by the conference announcement [71] and its accompanying full-length technical report. It essentially explains how if one changes the venue of application from integer arithmetic to a computer's floating point arithmetic environment, then Theorem 5's $IS_D(A)$ formalism can formally recognize each of the computer operations of addition, multiplication, subtraction and division as total functions among computerized representations of real numbers. In particular, [71] views a real number as an ordered pair of bit-strings for representing the mantissa and exponent, and it employs the usual computer hardware convention of rounding a real number after an arithmetic operation so that the outputted mantissa will have a bit-length no larger than the maximum of the lengths of the two input mantissas. This type of formalization of floating point arithmetic, described more fully in [71], is useful for reminding ourselves that there do exist certain *particularized respects* in which a self-justifying axiom system can recognize multiplication as a mathematically total function. Such applications do not change the fact that $IS_D(A)$ *cannot* recognize *integer* multiplication as a total function — and that our year-2002 paper [68] showed that no reasonable self-justifying axiom system can overcome this obstacle. However, it does show that there are certain types of computerized *real-valued cousins of integer* multiplication that $IS_D(A)$ can recognize as formally total.

§6. The Inconsistency Property of $NS_D^{k,m}(A)$. This section will have two parts. Section 6.1 will explain the intuitive reason that the formalisms $IS_D(A)$ and $NS_D^{k,m}(A)$ have contrasting consistency properties. Section 6.2 will then provide a formal proof that $NS_D^{k,m}(A)$ is inconsistent for most inputs of (k, m, D, A) .

6.1. The Intuitive Structural Difference Between the $IS_D(A)$ and $NS_D^{k,m}(A)$ Formalisms. Both the sentence $\Upsilon(k, m)$ and the formal system $NS_D^{k,m}(A)$ were

defined in Section 3. Thus, Equation (9) defined $\Upsilon(k, m)$ to be the Σ_1^* sentence “ $\exists z \text{Log}^k(z) \geq \overline{m}$ ”. The Definition 3 of Section 3 gave $\text{NS}_D^{k,m}(A)$ an identical definition as $\text{IS}_D(A)$ except for the following changes:

- a: $\text{NS}_D^{k,m}(A)$ ’s Group-zero schema contained the one additional axiom sentence of $\Upsilon(k, m)$.
- b: $\text{NS}_D^{k,m}(A)$ ’s Group-3 axiom had an identical structure as $\text{IS}_D(A)$ ’s Group-3 axiom — except that $\text{NS}_D^{k,m}(A)$ ’s Group-3 axiom was sensitive to the change that was made by Item (a) above. Thus, the word “this” when $\text{NS}_D^{k,m}(A)$ ’s Group-3 axiom declared “*This system is consistent*” had a connotation that was sensitive to the fact that the construct “this” referred to a system that includes the axiom $\Upsilon(k, m)$ within its scope.

A key point is that if one compares the formalisms $\text{IS}_D(A)$ and $\text{NS}_D^{k,m}(A)$, both are capable of proving the statement $\Upsilon(k, m)$ as a theorem. Their only significant difference is that the Group-3 axiom under $\text{NS}_D^{k,m}(A)$ makes a stronger statement because its usage of the word “this” has a connotation which treats $\Upsilon(k, m)$ as being a formal axiom (*in addition to* being a theorem).

This section will prove the counter-intuitive result that Theorem 5’s consistency preservation property collapses under the $\text{NS}_D^{k,m}(\bullet)$ mapping function, for most values of (k, m, D) . Essentially, this distinction will arise because the changed scope of the word “this” under $\text{NS}_D^{k,m}(A)$ ’s Group-3 axiom will prevent $\text{NS}_D^{k,m}(A)$ from satisfying Part-ii of Section 1’s definition for introspective unification.

Underlying Nature of this Distinction: Let $\zeta(p)$ denote a function that maps an initial proof p from the union of the Group 0, 1 and 2 axioms of $\text{NS}_D^{k,m}(A)$ onto the shortest proof of the same theorem from the Group 0, 1 and 2 axioms of $\text{IS}_D(A)$. Also, the Definition 2 (from Section 3) had defined the symbol 2_k^m . Using this notation, it is apparent there exists a fixed constant $C > 2_k^m$ such that C assuredly bounds the difference between the proof-lengths of p and $\zeta(p)$ when the deduction method D used by the $\text{NS}_D^{k,m}(A)$ and $\text{IS}_D(A)$ formalisms is the $\text{Tab}-U_1^*$ deduction method.

In Proof Theory, two axiom systems typically have similar properties when their proof lengths differ by only an additive constant. However, what is unusual about the particular systems $\text{IS}_D(A)$ and $\text{NS}_D^{k,m}(A)$ is that:

1. The fixed constant $C > 2_k^m$ separating these two proof lengths is an enormous quantity.
2. As noted by the opening paragraph of this section, the meaning of the word “this” in the Group-3 axiom sentences of $\text{IS}_D(A)$ and $\text{NS}_D^{k,m}(A)$ is affected by Item (1)’s distinction.

The complicated interaction between Items (1) and (2) is the reason why $\text{IS}_D(\bullet)$ is consistency preserving but $\text{NS}_D^{k,m}(\bullet)$ is not. (In particular, a Kleene-like Group-3

axiom has a different connotation when it refers to a short proof, referencing itself, than when it refers to a long such proof.) The remainder of this section will prove $NS_D^{k,m}(A)$ is inconsistent for most inputs (k, m, D, A) .

6.2. Formal Proof of $NS_D^{k,m}(A)$'s Inconsistency Property. Let $Cons_\alpha(n)$ be the “finistic consistency statement” which declares that there is no proof of $0=1$ from the axiom system α whose length is less than n . Both Friedman and Pudlák [14, 40, 43, 44] have observed that for some constant $\epsilon > 0$, most consistent systems α require a bit-length at least as large as $O(n^\epsilon)$ to prove $Cons_\alpha(n)$. These results do not apply to all self-justifying axiom systems α that recognize addition but not multiplication as a total function. For instance, consider the possibility that $n = 2^b$, D denotes $Tab-U_1^*$ -List deduction, $\alpha = IS_D(A)$ and we encode $Cons_\alpha(2_s^b)$ as: “There exists no $Tab-U_1^*$ -List proof p of $0=1$ from $IS_D(A)$ satisfying the inequality $Log^s(p) < b$ ”. Then $IS_D(A)$ is consistent, and it can use its Group-3 axiom to obtain a significantly shorter than $O(n^\epsilon)$ sized proof of $Cons_\alpha(2_s^b)$. However, $NS_D^{k,m}(A)$ will have very different properties: We will formally prove that $NS_D^{k,m}(A)$ is inconsistent by employing a finistic consistency statement. Our discussion will employ the following notation:

- i: The symbol PA will denote Peano Arithmetic. The symbol $PA+$ will denote the minor extension Peano Arithmetic that contains further function symbols for Section 2's U-Grounding language. (The footnote ⁷ explains why it is *actually permissible* for the sub-component $PA+$ of the $IS(PA+)$ and $NS_D^{k,m}(PA+)$ systems to employ *internally within itself* a multiplication function symbol, *provided that it is remembered* that the “ Π_1^+ theorems” of $PA+$, used by Equation (5)'s Group-2 axioms, are forbidden from containing appearances of the multiplication function symbol.)
- ii: $Subst(g, h)$ will denote Gödel's well-known Δ_0^* substitution formula — which yields TRUE when g is an encoding of a formula and h is an encoding of a sentence that replaces all occurrence of free variables in g with the Gödel-encoded term $\overline{g}$.
- iii: $\bar{U}$ will denote the Gödel diagonalization sentence which states “There exists no proof (using deduction method D) of *this sentence* from the axiom system $NS_D^{k,m}(PA+)$ ”.

⁷The point is that Equation (5)'s formalization of the Group-2 axiom-scheme *permits* its axioms to use a theorem Φ from the axiom system $PA+$ to transfer axiomatic information to the axiom system of $IS(PA+)$ or to $NS_D^{k,m}(PA+)$ *only when* Φ *does not violate the Π_1^+ condition* by containing Peano Arithmetic's multiplication function symbol. However, the formalism $PA+$ *is certainly permitted* to prove an intermediate theorem Ψ , *possessing some multiplication function symbols*, that is subsequently applied to prove a further Π_1^+ theorem Φ , which will be employed by Equation (5)'s Group-2 axiom. (Such an action is permitted *if* Φ (*unlike* Ψ) *does not contain multiplication function symbols under the paradigm we describe.*)

iv: $\mathcal{U}(s, b)$ will denote the variant of Item (iii)'s Gödel sentence that applies to proofs whose size is bounded by 2_s^b . It will formally state: "There exists no proof p (using deduction method D) of *this sentence* from the axiom system $\text{NS}_D^{k,m}(\text{PA}+)$ which satisfies $p < 2_s^b$ ".

To achieve our main results, it will be necessary that $\mathcal{U}(s, b)$ have a Π_1^* encoding. It is easy to accomplish this. Thus, let $\Gamma(g)$ denote the Π_1^* formula defined by Equation (24), and let n denote (24)'s Gödel number. Then $\mathcal{U}(s, b)$ is formally defined to be the Π_1^* sentence $\Gamma(\overline{n})$.

$$(24) \quad \forall p \forall h \{ [\text{Subst}(g, h) \wedge \text{Log}^s(p) < \overline{b}] \rightarrow \neg \text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(h, p) \}$$

DEFINITION 6. The symbol $\text{CheckProof}(t, p, q)$ will denote a Δ_0^* formula indicating that the condition $\text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(t, p)$ is satisfied and that q represents a proof (using deduction method D) of Equation (25)'s statement from the axiom system $\text{NS}_D^{k,m}(\text{PA}+)$.

$$(25) \quad \exists p \text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(\overline{t}, p)$$

Also, $F(t, p)$ will denote the function, which given an input of (t, p) satisfying $\text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(t, p)$, will produce the minimal integer q satisfying $\text{CheckProof}(t, p, q)$.

LEMMA 3. *There exists a constant $C > 0$, such that if $q = F(t, p)$ then Definition 6's function F will guarantee that $\text{Log}(q) < O\{\lceil \text{Log}(p) \rceil^C\}$.*

Proof Sketch: The prior literature [1, 4, 8, 14, 20, 38, 40, 41, 50, 61, 64] has illustrated several examples of a constant C , a Gödel-encoding method M and an axiom system α such that if p is a M -encoded proof using α 's axioms then there will exist a second proof q satisfying $\text{Log}(q) < O\{\lceil \text{Log}(p) \rceil^C\}$ such that q is a proof that p is a M -encoded proof using α 's axioms. An analogous construction will verify Lemma 3's validity. In this case $\text{NS}_D^{k,m}(\text{PA}+)$ is the employed axiom system, and Appendix A formalizes the Gödel-encoding method. $\square$

COROLLARY 1. *There exists a constant $a_0 > 0$, such that if $a > a_0$ and if $p < 2_{a-1}^b$ is a proof of an arbitrary theorem t from the axiom system $\text{NS}_D^{k,m}(\text{PA}+)$ then $F(p, t) < 2_a^b$.*

PROOF. An immediate consequence of Lemma 3. $\dashv$

LEMMA 4. *There exists a constant $a_0 > 0$, such that if $a > a_0$ then for any constant b , Equation (26)'s Π_1^* sentence is both true and provable from Peano Arithmetic.*

$$\forall p \forall n \exists q < n \{ [\text{Log}^{a-1}p < \overline{b} \wedge \text{Log}^a n \geq \overline{b} \wedge \text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(\lceil \mathcal{U}(a-1, b) \rceil, p)]$$

$$(26) \quad \rightarrow \text{CheckProof}(\lceil \mathcal{U}(a-1, b) \rceil, p, q) \}$$

PROOF. Equation (26)'s statement is a formalistic Π_1^* assertion whose logical content is identical to Corollary 1's statement — for the special case where t represents the sentence $\mathcal{U}(a-1, b)$. Therefore, the logical validity of (26) is an immediate consequence of Corollary 1. Moreover, Peano Arithmetic can verify (26)'s statement because it can formalize the proof of Corollary 1. $\dashv$

THEOREM 6. *For simplicity, let us assume D denotes the Tab- U_1^* -List deduction method (although this theorem will actually generalize for all cut-free deduction methods such as semantic tableaux or Herbrand deduction). For each such deduction method D , there will exist a constant k_0 such that for every integer $k > k_0$, the axiom system $NS_D^{k,m}(PA+)$ is inconsistent.*

On the Significance of Theorem 6 : There are several stronger versions of Theorem 6 than the variant above. In essence, we will focus on proving a relatively weak version of this proposition because its proof is pleasantly shorter. The Remark 8, at the end of this section, lists some stronger versions of Theorem 6. The contrast between their negative results and Theorem 5's positive result will be even sharper than that for the particular variant proven below:

PROOF. Our proof of Theorem 6 will have two parts. First, we will construct a proof W of the Gödel sentence $\mathcal{U}(k-1, m)$ from the axiom system $NS_D^{k,m}(PA+)$. Then we will analyze the length of W 's proof and show it is short enough to force $NS_D^{k,m}(PA+)$ to become inconsistent.

Description of a 7-step Tab- U_1^* -List Proof of $\mathcal{U}(k-1, m)$ (called “W”): Lemma 4 had indicated that for any ordered pair (a, b) where $a > a_0$, Peano Arithmetic can prove Equation (26)'s Π_1^* sentence. Since $NS_D^{k,m}(PA+)$'s Group-2 scheme allows $NS_D^{k,m}(PA+)$ to prove every Π_1^* theorem of Peano Arithmetic, it follows that $NS_D^{k,m}(PA+)$ can also prove (26). Applying this result in the particular case where $k = a$, $m = b$ and k is presumed to be larger than a_0 , we obtain $NS_D^{k,m}(PA+)$ can certainly prove the statement:

$$\begin{aligned} \forall p \forall n \exists q < n \{ [\text{Log}^{k-1} p < \overline{m} \wedge \text{Log}^k n \geq \overline{m} \wedge \text{Prf}_{NS_D^{k,m}(PA+)}(\lceil \mathcal{U}(k-1, m) \rceil, p)] \\ (27) \quad \rightarrow \text{CheckProof}(\lceil \mathcal{U}(k-1, m) \rceil, p, q) \} \end{aligned}$$

Let us next recall that $\text{Pair}(x, y)$ denotes a Δ_0^* formula that is satisfied precisely when x is the Gödel number of a Π_1^* sentence and y is the Gödel number of the sentence that represents the negation of x 's formal statement. Under this notation convention, Equation (28) is certainly a logically valid statement. Moreover, $NS_D^{k,m}(PA+)$ must be able to prove the validity of Equation (28) (because its Groups 0 and 1 axioms endow it with a capacity to prove every Δ_0^* sentence that is valid under the standard model of the natural numbers.)

$$(28) \quad \text{Pair}(\lceil \mathcal{U}(k-1, m) \rceil, \lceil \neg \mathcal{U}(k-1, m) \rceil)$$

The hypothesis of Theorem 6 had focused on $\text{Tab}-U_1^*$ -List deduction (essentially for the sake of simplifying the current proof). This form of deduction consists of a sequence of semantic tableaux proofs, where each proof establishes the validity of some Π_1^* or Σ_1^* sentence that is allowed to be used in the subsequent further steps of the $\text{Tab}-U_1^*$ -List proof. In such a context, Equations (27) and (28) will represent the first two intermediate results that our $\text{Tab}-U_1^*$ -List proof shall deduce.

Let us next recall that $\text{NS}_D^{k,m}(\text{PA}+)$'s Group-3 axiom is identical to Equation (6)'s formulation of $\text{IS}_D(A)$'s Group-3 axiom, except that the relevant subscripts are now changed from $\text{IS}_D(A)$ to $\text{NS}_D^{k,m}(\text{PA}+)$. Thus, Equation (29) formalizes $\text{NS}_D^{k,m}(\text{PA}+)$'s Group-3 axiom:

(29)

$$\forall x \forall y \forall p \forall q \neg [\text{Pair}(x, y) \wedge \text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(x, p) \wedge \text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(y, q)]$$

The third step in our proof W will establish the validity of Equation (30) as an immediate consequence of the combination of Equations (28) and (29).

(30)

$$\forall p \forall q \neg \{ \text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(\lceil \mathcal{U}(k-1, m) \rceil, p) \wedge \text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(\lceil \neg \mathcal{U}(k-1, m) \rceil, q) \}$$

The fourth step of the proof W is mostly notational. The definitions of CheckProof and $\mathcal{U}(k-1, m)$ imply that $\text{CheckProof}(\lceil \mathcal{U}(k-1, m) \rceil, p, q)$ is logically equivalent to Equation (30)'s curly bracket expression. Therefore, a fourth tableaux proof can derive (31) as a consequence from (30).

(31)

$$\forall p \forall q \neg \text{CheckProof}(\lceil \mathcal{U}(k-1, m) \rceil, p, q)$$

Combining the intermediate results from Equations (27) and (31), our fifth semantic tableaux proof will then derive the result of:

(32)

$$\forall p \forall n \neg \{ \text{Log}^{k-1} p < \overline{m} \wedge \text{Log}^k n \geq \overline{m} \wedge \text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(\lceil \mathcal{U}(k-1, m) \rceil, p) \}$$

Since the axiom $\Upsilon(k, m)$ (defined in Equation (9)) indicates the existence of an integer n satisfying $\text{Log}^k n \geq \overline{m}$, a sixth semantic tableaux proof can now deduce (33) from (32).

(33)

$$\forall p \neg \{ \text{Log}^{k-1} p < \overline{m} \wedge \text{Prf}_{\text{NS}_D^{k,m}(\text{PA}+)}(\lceil \mathcal{U}(k-1, m) \rceil, p) \}$$

Equation (33) is clearly equivalent to $\mathcal{U}(k-1, m)$'s Gödel sentence. It is therefore routine for a seventh tableaux proof to use Equation (33)'s intermediate result to arrive at the deduction of $\mathcal{U}(k-1, m)$. The $\text{Tab}-U_1^*$ proof, consisting of these seven semantic tableaux subcomponents, is defined to be the formal proof called W .

Finishing the Proof of Theorem 6 by Analyzing the Formal Length of W 's Proof: The critical difference between the axioms systems $\text{IS}_D(A)$ and $\text{NS}_D^{k,m}(\text{PA}+)$ is that the latter formalism contained Equation (9)'s added Group-zero axiom, which we denoted as $\Upsilon(k, m)$. This axiom is technically unnecessary for deducing Equation (33) from (32) (because if the axiom $\Upsilon(k, m)$ was absent then our proof methodology

could still derive $\Upsilon(k, m)$ as a theorem). However, the point is that if $\Upsilon(k, m)$ is a formal axiom, then the resulting proof, deriving (33) from (32), is sharply shorter. As a result of this proof compression, it is quite routine to obtain that there exists a constant k_0 such that every ordered pair (k, m) with $k > k_0$ will automatically satisfy the condition that the proof W (constructed above) has a Gödel number less than 2_{k-1}^m .

The latter observation will now enable us to complete the proof of Theorem 6. This is because it shows that that $NS_D^{k,m}(PA+)$ can prove the Gödel sentence $\mathcal{U}(k-1, m)$ with a proof whose length is so short — that the very existence of such a proof is precluded by $\mathcal{U}(k-1, m)$'s formal statement. Hence, the axiom system $NS_D^{k,m}(PA+)$ is inconsistent, as Theorem 6 had claimed. $\dashv$

REMARK 8. (*concerning stronger versions of Theorem 6*): There are several stronger versions of Theorem 6. They arise because as the values for k and m increase, this chapter's proof engine gains new applications. Among these stronger versions of Theorem 6 are:

1. A more sophisticated variant which indicates that if one increases the values of (k, m) by a small amount then $NS_D^{k,m}(PA+)$ will also be inconsistent when D denotes semantic tableaux, Herbrand deduction or the cut-free version of the sequent calculus,
2. A variant of Theorem 6 which states $NS_D^{k,m}(A)$ remains inconsistent when A represents the empty axiom system (instead of $PA+$) and (k, m) is made sufficiently large. The footnote ⁸ explains why large enough values of (k, m) enable Theorem 6 to generalize in this direction.
3. A third type of generalization of Theorem 6 where for each cut-free deduction method D there exists a finite-sized non-growth axiom system A and a pair of large enough integers (k, m) where a modified form of $NS_D^{k,m}(A)$, which drops the two Group-zero axioms that had declared addition and doubling were total functions, is inconsistent. Thus, this variant of $NS_D^{k,m}(A)$ is inconsistent even though it fails to recognize the assured existence of any integer larger than 2_k^m .

Main Perspective: The broader significance of Theorems 5 and 6 is difficult to appreciate until these two results are contrasted in detail. This is because the opposing results of these two theorems are quite counter-intuitive — especially when their added generalizations in Remarks 1, 6, 7 and 8 are also contrasted.

⁸The prior literature on weak axiom systems [20, 23, 34, 41, 52, 64] has noted that a good part of the force of the Principle Induction applies to the set of integers lying within the ranges of well-defined Definable Cuts, even when an axiom system does not recognize the formal validity of the Principle Induction. The proof of Theorem 6 can be modified to gainfully employ this fact when (k, m) represents a large enough pair of fixed constants. The net effect of this observation is that Theorem 6 will remain valid in the extremal case where A represents the empty axiom system — provided the threshold values for k and m are appropriately increased.

Contrast Between Σ_1^* and Π_1^* Sentences: Let us recall that Section 3’s definition of $NS_D^{k,m}(A)$ involved adding a logically valid Σ_1^* sentence to $IS_D(A)$, called $\Upsilon(k, m)$, and then adjusting the Group-3 axiom to reflect this change. For any fixed sentence Ψ , let $N_D(A, \Psi)$ denote the analog of $NS_D^{k,m}(A)$ where the sentence added to Group-zero is Ψ (instead of $\Upsilon(k, m)$), and the Group-3 “I am consistent” statement is accordingly adjusted again. Although Theorem 6 showed $NS_D^{k,m}(A)$ was typically inconsistent, there is no analog of this inconsistency when Ψ is a logically valid Π_1^* sentence. Instead, Theorem 5’s *opposing positive result* can be generalized to show that $N_D(A, \Psi)$ is always consistent when Ψ and all A ’s Π_1^* theorems are valid in the standard model.

§7. Concluding Remarks. The two main results of this article are the opposing positive and negative effects of Theorems 5 and 6. This subject matter should generate scholarly interest — even if its main lasting significance is seen as only defining more precisely the exact mathematical borderline that separates the Second Incompleteness Theorem from its boundary-case exceptions.

The $IS_D(A)$ axiom system of Theorem 5 is stronger than the $IS(A)$ system of our prior article [67] in two respects. These are that 1) $IS_D(A)$ uses a Level(1) rather than Level(0-) definition of self-consistency and 2) its mode of deduction replaces the semantic tableaux methodology with the more flexible Tab- U_1^* -List approach.

As Remark 1 had noted, Theorem 5’s consistency property can be strengthened by extending it from Tab- U_1^* to Tab- Q_1^* deduction. In essence, we opted for presenting a simplified version of Theorem 5 because it was preferable to broaden the audience for this paper through Section 5’s shorter form of presentation, rather than to offer its strongest version.

The opposing result of Theorem 6 was that $IS_D(A)$ ’s cousin, called $NS_D^{k,m}(A)$, is inconsistent. This contrast is very surprising because the main difference between $IS_D(A)$ and $NS_D^{k,m}(A)$ is the connotation their respective Group-3 axioms associate with the word “this”.

A more detailed commentary about $IS_D(A)$ ’s significance has been provided by Remarks 2-4 at the end of Section 5.1. These remarks discuss several further respects in which $IS_D(A)$ is a close-to-maximal variant of introspective formalism, whose additional strengthening is not feasible.

Also, Remark 2 explained how Section 5’s methodologies significantly simplifies the proof techniques from [67], in addition to providing a stronger and more precise form of result.

Before closing, it is also helpful to recall that the definitions for Σ_1^* and Π_1^* sentences had allowed for an arbitrary combination of bounded quantifiers to appear to the right of the unbounded quantifiers of a Σ_1^* and Π_1^* sentence. This condition is called the unlimited-bounded-quantifier property. If our formalism had not incorporated this level

of generality into the $\text{Tab}-U_1^*$ -List deduction method, then the significance of Theorem 5's $\text{IS}_D(A)$ axiom system could perhaps be interpreted as a much narrowed and very restricted form of result.

However in a context where [70] has established that the Second Incompleteness Theorem applies to both $\text{Tab}-\Sigma_2^*$ and $\text{Tab}-\Pi_2^*$ deduction, Theorem 5 can be proven to be a very close to maximal result. This is because $\text{Tab}-U_1^*$ -List's modus ponens rule is essentially the next weaker level of deduction available. It applies to *every* Σ_1^* and Π_1^* sentence satisfying the previous paragraph's unlimited-bounded-quantifier property, *rather than to* a narrower subset of this generic class (i.e. see footnote ⁹).

Thus, the $\text{IS}_D(A)$ system provides an unusual combination of qualities. It is unconventional firstly because $\text{IS}_D(A)$ does not recognize *integer-based* multiplication as a total function — a drawback whose consequences simply cannot be easily ignored — *even when Remark 7 noted* that $\text{IS}_D(A)$ can recognize floating point multiplication as a formally total function. Also, $\text{IS}_D(A)$'s Group-3 axiom employs an unusual form of restricted modus ponens principle. One reason $\text{IS}_D(A)$ should stimulate scholarly interest, despite its clearly evident partial disadvantages, is because it possesses a type of limited (and certainly not full) understanding of its own self-consistency — under a definition of consistency that is applicable to the previous paragraph's Level-1 styled form of deduction.

The main reason for interest in $\text{IS}_D(A)$ is, of course, because it is always helpful to consider unconventional boundary-case paradigms, to help sharpen one's understanding of great seminal theorems, such as for instance Gödel's historic incompleteness result. Our discussion has deliberately avoided speculations about the fuller epistemological implications of self-justifying axiom systems, similar to $\text{IS}_D(A)$, because such implications are very complex — especially when all self-justifying systems which affirm their own Level(0-) semantic tableaux consistencies *are inherently unable* [68] to prove multiplication is a total function *among integers*. Instead, our goal has been to sharpen the scholarly understanding of the Second Incompleteness Theorem's meaning, by having the contrasting positive and negative results of Theorems 5 and 6 help identify the mathematical threshold levels where this incompleteness theorem formally takes hold.

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⁹For instance, Theorem 5's significance would be much narrowed if for some fixed constant K , it had required that no more than K alternations of bounded universal and existential quantifiers can lie to the right of the unbounded quantifiers of a Σ_1^* and Π_1^* sentence.

Appendix A: Summary of Gödel Encoding Method. This appendix describes our formal method for encoding Semantic Tableaux Proof trees and Tab- $\mathfrak{R}$ -List proofs as Gödel numbers. Our encoding scheme will be roughly analogous to the “natural B-adic” encoding methods of Hájek-Pudlák [20] and Wilkie-Paris [64] — insofar as the number of utilized bits to encode a semantic object will be approximately proportional to the length of such an expression written by hand. This appendix will be somewhat unnecessary because a reader should be able to appreciate the gist of our article without examining a description of our Gödel encoding method. However in a context where Sections 2–6 often allude to the “Gödel encoding” of a syntactic object, it is clearly helpful to summarize our encoding method.

Our scheme for encoding a semantic tableaux proof will use the following 24 language symbols:

1. The standard connective symbols of $\wedge$, $\vee$, $\neg$, $\rightarrow$, $\forall$ and $\exists$.
2. Two left and two right parenthesis symbols denoted as: $($, $)$, $($ and $)$.
3. Two symbols to represent the special constants of “0” and “1”.
4. Eight function symbols for representing for representing the eight formal U-grounding functions of Addition, Doubling, Subtraction, Division, Logarithm, etc.
5. The relation symbols of “=” and “ $\leq$ ”.
6. The symbol $\hat{V}$ for designating the presence of a basic variable v in a logical sentence.
7. The symbol $\hat{U}$ for designating the presence of a parameter constant u in a logical sentence (which is produced by Section 2’s deduction rules 5 and 6 for eliminating existential quantifiers).

Define a byte to be an unit consisting of six bits. We will think of a proof as either comprising a sequence of bytes (or being an integer written in base 64). Each of the 24 symbols (above) will be given some unique 6-bit code, ranging between 32 and 55. Our method for representing the presence of the i -th variable v_i will be to encode it as a string of $\lceil \log_{32}(i+1) \rceil + 1$ bytes, where the first byte is the “ $\hat{V}$ ” symbol and the remaining bytes encode i as a base-32 number, with the convention that the lead bit in each byte’s 6-bit sequence is “0”. The same convention will be used to denote the presence of the i -th parameter u_i except its first byte will be the “ $\hat{U}$ ” symbol.

Our notation has employed *two types* of parenthesis symbols because the first pair of parenthesis symbols will have their usual meaning in punctuating a mathematical sentence, whereas the latter pair of symbols $($ and $)$ will *separate* the individual sentences in a Semantic Tableaux proof tree. For example, consider a tree which stores 1) the sentence ψ_1 as its root, 2) the sentences ψ_2 and ψ_3 as the root’s children, and 3) ψ_4 as the child of ψ_3 . There are several possible notation conventions for using the $($ and $)$ symbols to encode a Semantic Proof tree. Our encoding convention will be that ψ_i is an “ancestor” of ψ_j *if and only if* the range beginning with the parenthesis to ψ_i ’s immediate left and continuing to the matching right parenthesis includes ψ_j .

The example of our 4-node proof tree would thus be encoded as:

$$(34) \quad (\psi_1 (\psi_2) (\psi_3 (\psi_4))))$$

The preceding paragraph summarized our method for encoding semantic tableaux proofs. Its generalization for encoding of Tab- $\mathfrak{R}$ proofs is straightforward. Thus if $p_1, p_2, \dots, p_n$ collectively constitute a list of semantic tableaux proofs then the natural concatenation of their byte strings will be the corresponding Tab- $\mathfrak{R}$ proof.

This byte-styled encoding method is approximately analogous to what Wilkie-Paris [64] have called a *natural B-adic* encoding or a similar counterpart in the Hájek-Pudlák textbook [20]. Such compressed encodings are considered to be more meaningful and efficient than an uncompressed encoding method, using say a Prime Number decomposition scheme [30] (because the latter has an unnecessarily long bit-length). All our theorems would also be valid for uncompressed encoding methods. However, they are more meaningful when one uses an efficiently compressed B-adic encoding method.

Appendix B: Demonstration that semantic tableaux satisfies Part (B) of the definition of θ -Compactification. The proof that semantic tableaux satisfies the Part-B condition has been put into this appendix because it is similar to Section 5.2's proof of the Part-A condition.

The hypothesis of Theorem 2 had indicated that $\theta = \frac{1}{4}$. Since our discussion will employ a proof-by-contradiction, we will entertain the temporary hypothesis that the Part-B condition is false. Then there must exist a Π_1^* sentence Φ , that has a proof $p \leq \lceil \frac{a}{b} \rceil + 1$ from a Normed(a, b) axiom system α where Φ 's $App\forall(\frac{a}{p^\theta})$ interpretation is fallacious. Then p 's root will be a Σ_1^* sentence that represents the negation of Φ , and it will possess an $App\exists(\frac{a}{p^\theta})$ validity level.

Let Z denote the axiom system that is the union of this Σ_1^* representation of $\neg\Phi$ with α . The footnote ¹⁰ proves Z is a Normed($a, \frac{a}{p^\theta}$) system (by employing a minor variation of the formalism that Section 5.2 used in Part-A's proof.)

The footnote ¹¹ shows that the pair (p, Z) then must satisfy Lemma 1's hypothesis. Hence, Lemma 1 will ascertain that p must possess a positive branch.

¹⁰We can assume $\frac{a}{p^\theta} \geq b$ holds because the first sentence of Footnote 4's proof had proven the validity of the inequality: $p^\theta \cdot b \leq a$. The key point is that the combination of the fact that α is a Normed(a, b) system and the $\frac{a}{p^\theta} \geq b$ inequality imply α is also a Normed($a, \frac{a}{p^\theta}$) system. Since the Σ_1^* representation of $\neg\Phi$ possesses an $App\exists(\frac{a}{p^\theta})$ level of validity, this implies that the union of $\neg\Phi$ with α is another Normed($a, \frac{a}{p^\theta}$) system.

¹¹The combination of Equation (20), our assumption that $\theta = \frac{1}{4}$ and the equality $\text{Log}_2(a) - \text{Log}_2 \frac{a}{p^\theta} = \theta \cdot \text{Log}_2 p$ immediately imply that $\text{U-Height}(p) < \text{Log}_2(a) - \text{Log}_2 \frac{a}{p^\theta}$. The latter inequality shows that the pair (p, Z) satisfies Lemma 1's hypothesis.

We will now bring our proof-by-contradiction to a conclusion by employing Lemma 2. It indicates that such a positive branch of p cannot be closed. Hence, p cannot be a semantic tableaux proof (because one of its branches remains open). This observation brings our proof-by-contradiction to its desired end by contradicting the initial assumption that p was a proof of Φ . $\square$

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